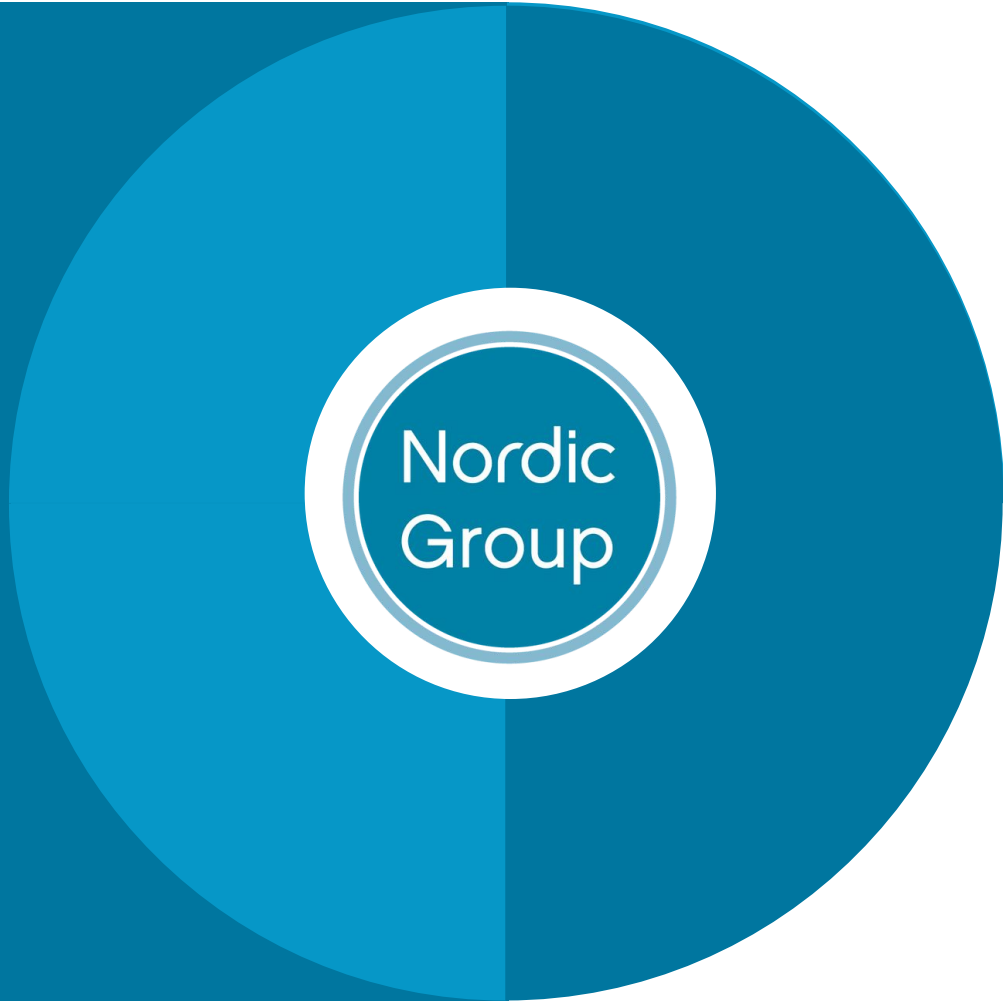
#HealthCareForTheFuture

Die Nordic Konferenz Funktionelle Diagnostik von Kinderwunsch bis Longevity



Chris Moore, CEO



Chris Moore

CEO Nordic Group,
Denmark

Spent 30 years working in the **functional/personalised medicine** segment.

Building the infrastructure to **shift healthcare** from reactive treatment to proactive, data-driven optimization.

A lifelong commitment to “biohacking”, building **“longitudinal health data”** to build resilience against disease.

Advisory Board member for the **Yale University Precision Wellness Initiative** and **UK Society for Longevity Medicine** International speaker on longevity and health.

The history



Nordic Laboratories

Founded 1997

Anne Catherine Færgemann

Chris Moore

1997-today


Nordic
Laboratories
lised Healthcare Starts
Right Test Combination


3/8
Oskar & Ollie
Oskar Moore
Business Dev & Strategic Advisor at Nordic. Head of Strategy at Functional Future
Oliver Moore
CEO At Functional Future, Head Of Copenhagen Office at Nordic.

Nordic Group 2026

A global company

Nordic Laboratories and Dnalife in 2026

Samples from +110 countries every month

We have 25 nationalities employed
Speak 21 languages



Nordic Group:
The Flywheel

- **Established 1997**: A group of companies delivering personalised medicine solutions.
- **Global Ecosystem**: Headquarters in Denmark, operating a strategically interconnected network of health companies.
- **Predictive, Not Reactive**: Leveraging comprehensive *longitudinal data* to shift the future of medicine from treating illness to optimizing health.

How Resistomap Works

resistomap



COLLECT SAMPLES

Wastewater / soil / water



EXTRACT eDNA

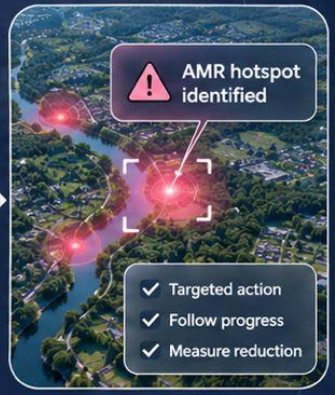


QUANTIFY RESISTANCE

Up to 384 genes



ANALYSE & COMPARE



TAKE ACTION

QUANTIFY → UNDERSTAND → ACT

Environmental antimicrobial resistance monitoring | From samples to actionable insights

Antibiotic resistance

Overview



But **1 in 6 deaths** are caused by bacterial infections

Antimicrobial Resistance (AMR) threaten **$100 Trillion in Economic Losses**

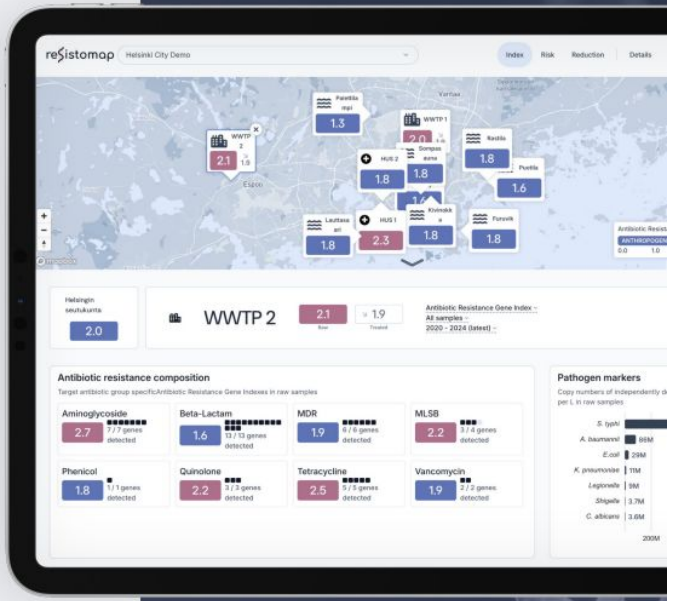
As biological threats escalate in pace from increased pollution, we believe data can provide us with solutions to protect planetary and human health

The solution

Giving governments, healthcare and industry the real-time intelligence they need to

Anticipate	Prevent	Respond to
✓	✓	✓

Bacterial threats



DSRUPTIVE

How DSruptive Works

Implantable NFC technology for Humans



Micro implant

A discreet, biocompatible device (3 × 12 mm)



Implanted

A quick, minimally invasive procedure



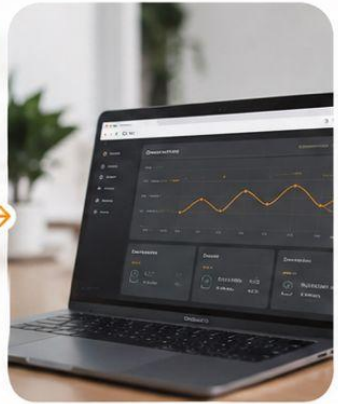
Scan with phone

Tap the implant with an NFC-enabled phone



Read temperature

On-demand, real-time temperature data



Connect & integrate

Secure data for health, identity and digital services

IMPLANT → SCAN → CONNECT

NFC technology | Body temperature monitoring | Secure identity & access | Integration with health services | Scalable platform

FUNCTIONAL FUTURE



Organic Extra Virgin Olive Oil



Organic Bone Broth Collagen



The Ferment



Organic Beef Bone Broth



Biodynamic Ginger Turmeric Elixir



Organic Organ Matrix Capsules



Organic Calcium Bone Matrix



Organic Beef Liver Capsules

Framing Health

We are all unique: Our health is determined by our **inherent differences** combined with our **lifestyles & environment.**

By combining and analysing information about our **genome**, with other diagnostic information, **patterns** can be identified that can help determine individual risk of developing disease; detect illness earlier; and, determine the most effective interventions to help improve our health, be they medicines, lifestyle choices, or even simple changes in diet.

The British National Health Service
https://www.england.nhs.uk/healthcare-science/personalisedmedicine/

Build Resilience

Precision Interventions: **test!**
genes, biomarkers, toxins

Employ dietary change, nutraceuticals, wearables - **build longitudinal data & extend healthspan**

We are complex beings: but the health landscape has for 100 years tried to protocol manage humans. It has failed. **Disease continues to grow, in tact with the spend in national health services**

The n=1 imperative

Hands up if you have done the following tests on yourself:



Genetics

Hands up if you have done the following tests on yourself:



Genetics



Biological Age

Hands up if you have done the following tests on yourself:



Genetics



Biological Age



Microbiome

Hands up if you have done the following tests on yourself:



Genetics



Biological Age



Microbiome



Biochemistry

Hormonal, Toxicity, or Neurological testing?

Hands up if you have done the following tests on yourself:



Genetics



Biological Age



Microbiome



Biochemistry

Hormonal, Toxicity, or Neurological testing?



The Question:

How many of you know that what you do is **OPTIMAL** for you?

Are you taking omega 3 fatty acids, or eating fish several times a week?

Do you know if it is affecting your fatty acid ratios? Improving your omega 3 status?

Fatty Acid bloodspot test

Inflammation balance

Cell membrane health

Omega-3 status



Nordic Laboratories

PATIENT: XXXXXXXXXXXXXXXXX
TEST NUMBER: T-NL-XXXXX (XXXXXXXXXX)
GENDER: XYZ
AGE: XX

COLLECTED: XX/XX/XXXX
RECEIVED: XX/XX/XXXX
TESTED: XX/XX/XXXX

TEST REF: TST-NL-XXXX
PRACTITIONER:
XXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXX

TEST NAME: Omega-3 / VitD / HbA1c (BloodSpot)

Omega-3 Index Complete Report

Your Omega-3 Index

Reference Range*: 2.8% - 15.4%

10.8%

YOUR LEVEL

Desirable Range: 8% - 12%

1% 2% 3% 4% 5% 6% 7% 8% 9% 10% 11% 12%

Fatty Acid bloodspot test

Inflammation balance

Cell membrane health

Omega-3 status



Nordic Laboratories

PATIENT: XXXXXXXXXXXXXXXX
TEST NUMBER: T-NL-XXXXX (XXXXXXXXXX)
GENDER: XYZ
AGE: XX

COLLECTED: XX/XX/XXXX
RECEIVED: XX/XX/XXXX
TESTED: XX/XX/XXXX

TEST REF: TST-NL-XXXX
PRACTITIONER:
XXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXX

TEST NAME: Omega-3 / VitD / HbA1c (BloodSpot)

Omega-3 Index Complete Report

Your Omega-3 Index

Reference Range*: 2.8% - 15.4%

6.10%

YOUR LEVEL

Desirable Range: 8% - 12%

1% 2% 3% 4% 5% 6% 7% 8% 9% 10% 11% 12%

Nordic Group

Fatty Acid bloodspot test

Inflammation balance

Cell membrane health

Omega-3 status



Nordic Laboratories

PATIENT: XXXXXXXXXXXXXXXXX
TEST NUMBER: T-NL-XXXXX (XXXXXXXXXX)
GENDER: XYZ
AGE: XX

COLLECTED: XX/XX/XXXX
RECEIVED: XX/XX/XXXX
TESTED: XX/XX/XXXX

TEST REF: TST-NL-XXXX
PRACTITIONER:
XXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXX

TEST NAME: Omega-3 / VitD / HbA1c (BloodSpot)

Omega-3 Index Complete Report

Your Omega-3 Index

Reference Range*: 2.8% - 15.4%

2.2%

YOUR LEVEL

Desirable Range: 8% - 12%

1% 2% 3% 4% 5% 6% 7% 8% 9% 10% 11% 12%

Nordic Group

Are you eating enough protein? Taking protein shakes?

Are you absorbing the amino acids?
Do you know your amino acid profile?

Amino Acid

Nutrient sufficiency

Detox support

Neurotransmitter building



3503 Bloodspot Amino Acids 20 Profile- Blood

Methodology: High Performance Liquid Chromotography

Ranges: Ages 13 and over.

	Results µmol/L	QUINTILE DISTRIBUTION (1st / 2nd / 3rd / 4th / 5th)	95% Reference Range
Essential Amino Acids			
Limiting Amino Acids			
1. Lysine	127	92 – 183	63 - 220
2. Methionine	13	12 – 28	10 - 33
3. Tryptophan	35	28 – 45	24 - 52
Branched Chain Amino Acids			
4. Isoleucine	44	35 – 77	28 - 96
5. Leucine	75	71 – 139	59 - 162
6. Valine	155	126 – 229	105 - 266
Other Essential Amino Acids			
7. Phenylalanine	49	43 – 72	37 - 86
8. Histidine	70	31 – 84	22 - 99
9. Threonine	96	67 – 143	54 - 169
Conditionally Essential Amino Acids			
10. Arginine	49	28 – 71	17 - 91
11. Taurine	183	145 – 245	124 - 282
12. Glycine	213	243 – 449	207 - 559
13. Serine	121	95 – 219	79 - 310

Nordic Group

Amino Acid

Nutrient sufficiency

Detox support

Neurotransmitter building

Good profile

3503 Bloodspot Amino Acids 20 Profile- Blood

Methodology: High Performance Liquid Chromotography

Ranges: Ages 13 and over.

	Results µmol/L	QUINTILE DISTRIBUTION 1st–5th	95% Reference Range
Essential Amino Acids			
Limiting Amino Acids			
1. Lysine	127	92 – 183	63 - 220
2. Methionine	13	12 – 28	10 - 33
3. Tryptophan	35	28 – 45	24 - 52
Branched Chain Amino Acids			
4. Isoleucine	44	35 – 77	28 - 96
5. Leucine	75	71 – 139	59 - 162
6. Valine	155	126 – 229	105 - 266
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Nordic Group

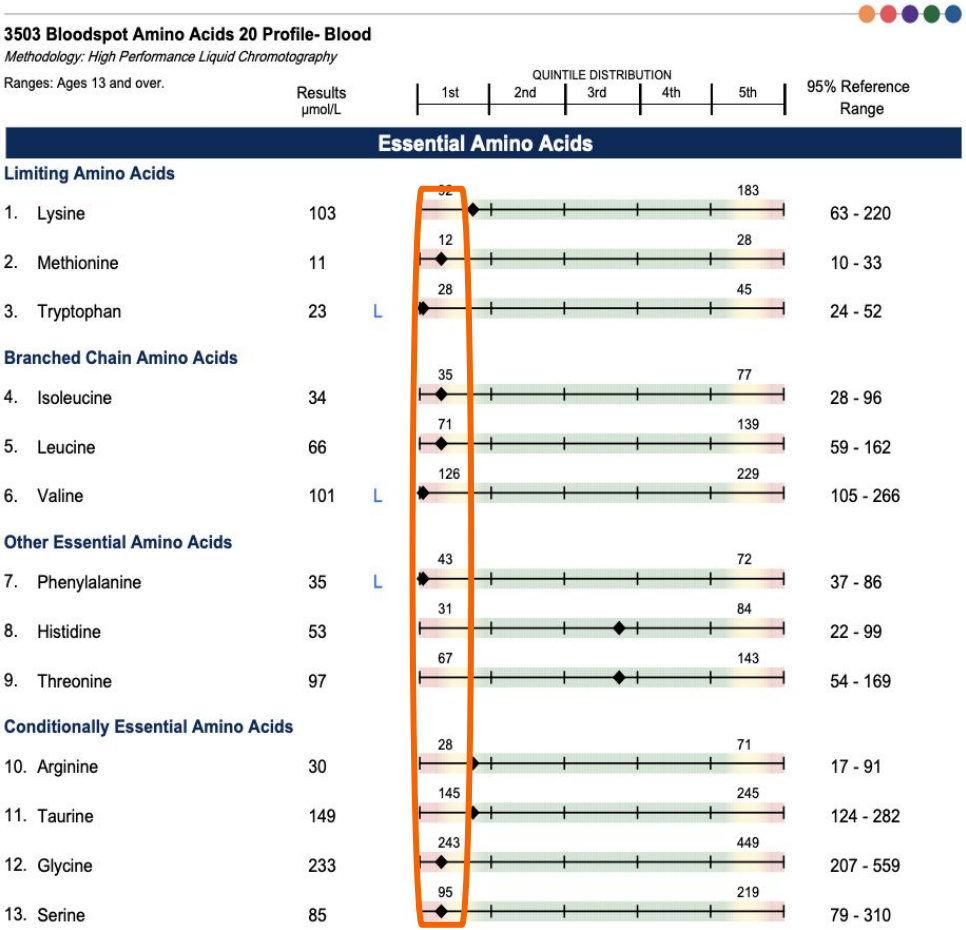
Amino Acid

Nutrient sufficiency

Detox support

Neurotransmitter building

Bad profile



3503 Bloodspot Amino Acids 20 Profile- Blood

Methodology: High Performance Liquid Chromotography

Ranges: Ages 13 and over.

	Results µmol/L		QUINTILE DISTRIBUTION 1st 2nd 3rd 4th 5th	95% Reference Range
Essential Amino Acids				
Limiting Amino Acids				
1. Lysine	103		183	63 - 220
2. Methionine	11		12 28	10 - 33
3. Tryptophan	23	L	28 45	24 - 52
Branched Chain Amino Acids				
4. Isoleucine	34		35 77	28 - 96
5. Leucine	66		71 139	59 - 162
6. Valine	101	L	126 229	105 - 266
Other Essential Amino Acids				
7. Phenylalanine	35	L	43 72	37 - 86
8. Histidine	53		31 84	22 - 99
9. Threonine	97		67 143	54 - 169
Conditionally Essential Amino Acids				
10. Arginine	30		28 71	17 - 91
11. Taurine	149		145 245	124 - 282
12. Glycine	233		243 449	207 - 559
13. Serine	85		95 219	79 - 310

Nordic Group

Amino Acid

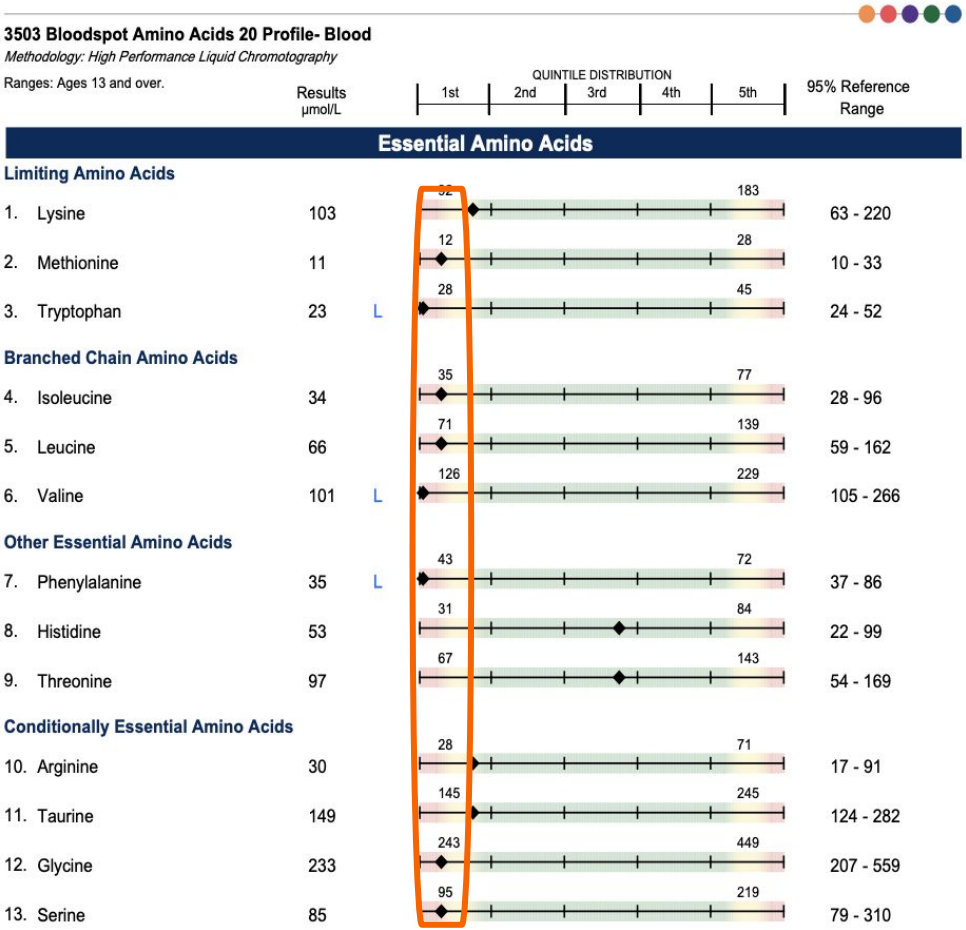
Nutrient sufficiency

Detox support

Neurotransmitter building

Bad profile

Consider protein intake and whether there are absorption issues



3503 Bloodspot Amino Acids 20 Profile- Blood

Methodology: High Performance Liquid Chromotography

Ranges: Ages 13 and over.

	Results µmol/L		QUINTILE DISTRIBUTION 1st 2nd 3rd 4th 5th	95% Reference Range
Essential Amino Acids				
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Nordic Group

Are you getting enough vitamin D from sun, diet or supplements?

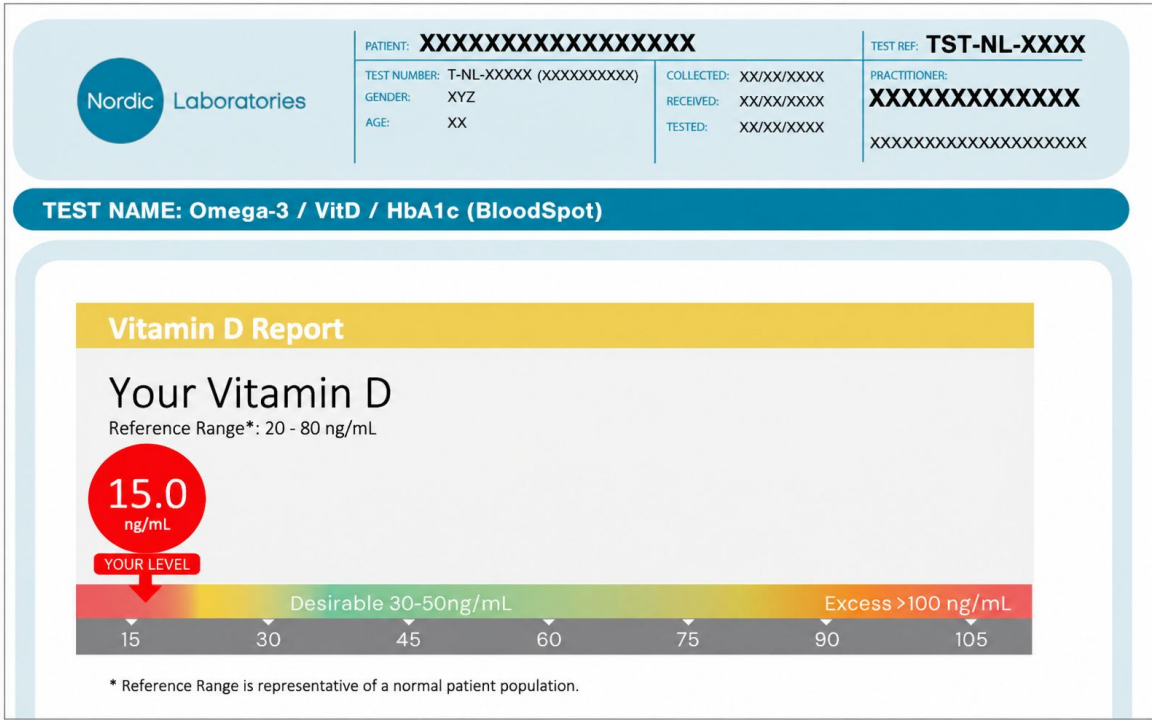
Do you know you vitamin D status?

Vitamin D

Immune regulation

Bone health

Hormone support

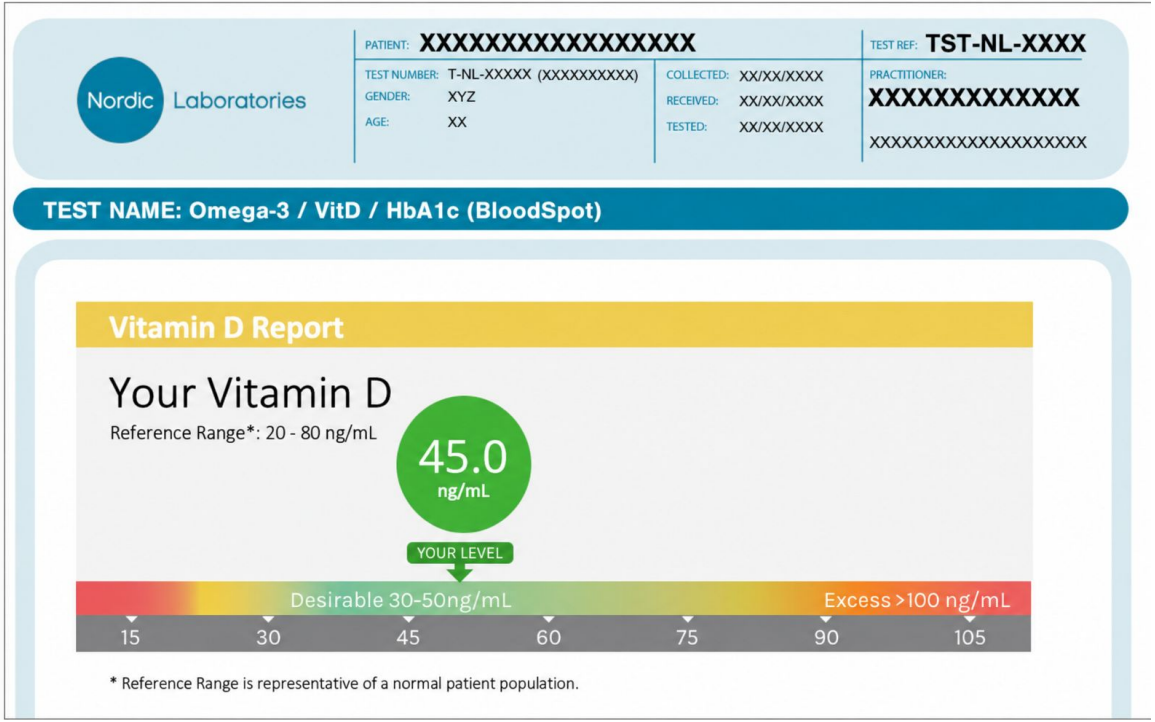


Vitamin D

Immune regulation

Bone health

Hormone support



If AI could help you live the perfect life, in good health and happiness,

but you have to follow **EVERY piece of advice from the AI bot**

would you do it?



Imagine you are lying in a bed surrounded by your loved ones and in your final hours...



Close your eyes...and feel yourself in that bed

How old were you in that moment?

70’s?

80’s?

90’s?

100’s?

Older?!!!!

Did you pass away pain free and happy that your life was lived fully all the way through?


Are you playing tennis at 100?

Anyone die of any disease?

You didn’t have:

Cancer

You didn’t have:

Cancer

You didn’t have:

Chronic Pain

Lung Disease

You didn’t have:

Cancer

Chronic Pain

Lung Disease

Cancer

You didn’t have:

Chronic Pain

Infirmity

Lung Disease

Cancer

You didn’t have:

Chronic Pain

Diabetic Foot

Infirmity

CVD

Lung Disease

Cancer

You didn’t have:

Chronic Pain

Diabetic Foot

Infirmity

CVD Lung Disease **Arthritis** Cancer

You didn’t have:

Chronic Pain Diabetic Foot Infirmity

CVD Lung Disease Arthritis Cancer

You didn’t have:

Chronic Pain Diabetic Foot Infirmity **Dementia**

Condition	Lifetime Risk (%)	Avg. Age of Onset	Avg. Age of Death	Statistical Chance of Death
Cardiovascular Disease	40-50%	60-70	78–83	~1 in 2
Cancer (All types)	25-35%	65-72	72–76	~1 in 4
Dementia/Alzheimers	20-25%	80+	85–88	~1 in 10
Lung Disease (COPD/CA)	10–15%	55-65	70–75	~1 in 20
Diabetic Foot/Complications	19-34% (of Diabetics)	55–65	74–78	High (via sepsis/CVD)
Chronic Pain/Arthritis	20-35%	50-65	N/A*	Low direct / High indirect

academic.oup.com/eurheartj/article/37/42/3232/2536403
joint-research-centre.ec.europa.eu/jrc-news-and-updates/latest-eu-cancer-data-what-new-2025-12-17_en
thelancet.com/gbd

100’s?

100’s?
1.5% Chance of getting to 100

The risk is that too many of our final years are

not

spent in good health.

Longevity: Why testing genes, biomarkers & toxins mitigate disease and extend healthspan.

Health must be measured, we have the tools, and the measuring is continual.

“Longitudinal Health”

Genes vs environment

The trigger concept

If genes load the gun, the environment pulls the trigger.

Identify the environmental and biochemical factors that activate predispositions.

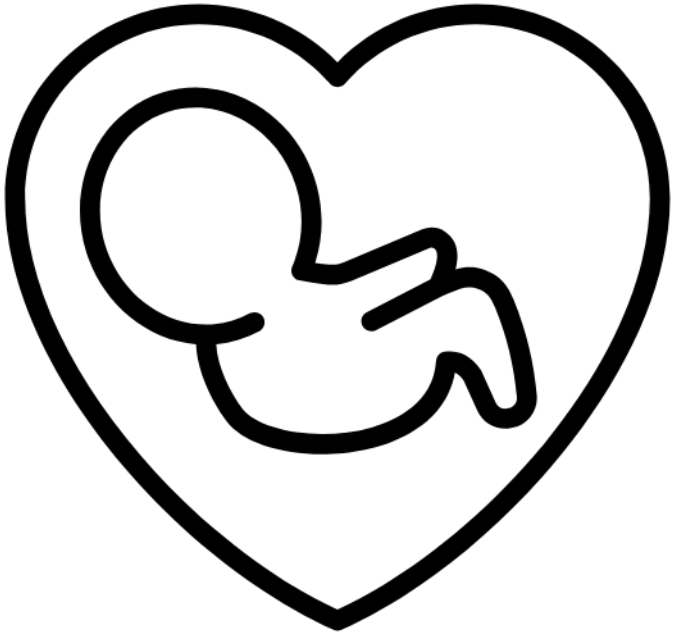
- Identifying SNPs (Single Nucleotide Polymorphisms)
- Mapping biochemical individuality
- Intervening before the "Trigger" is pulled



The prenatal foundation

The first 1000 days

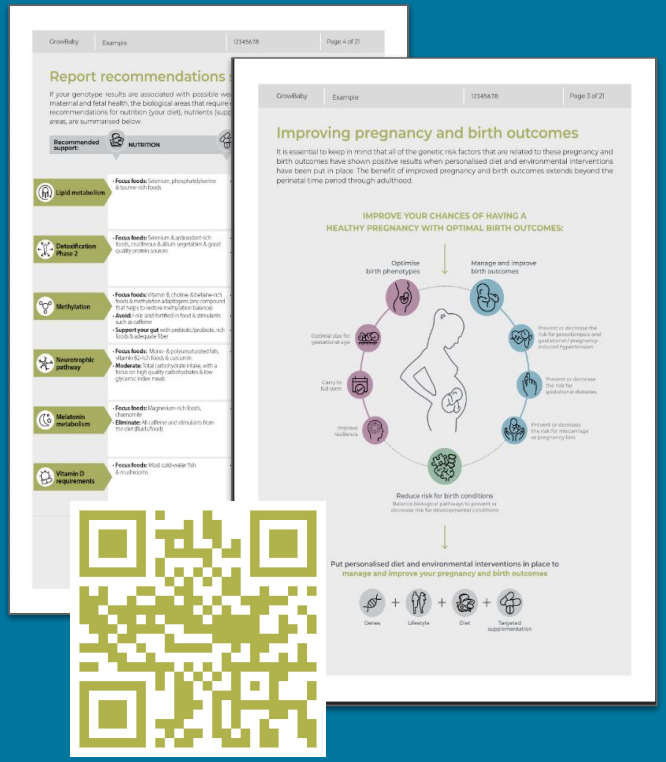
- Longevity is not an old-age strategy; it is a life-long evolution.
- Precise intervention begins before birth.
- By understanding maternal genetics and epigenetic influences, we can optimize fetal development and set the blueprint for a disease-free century of life.



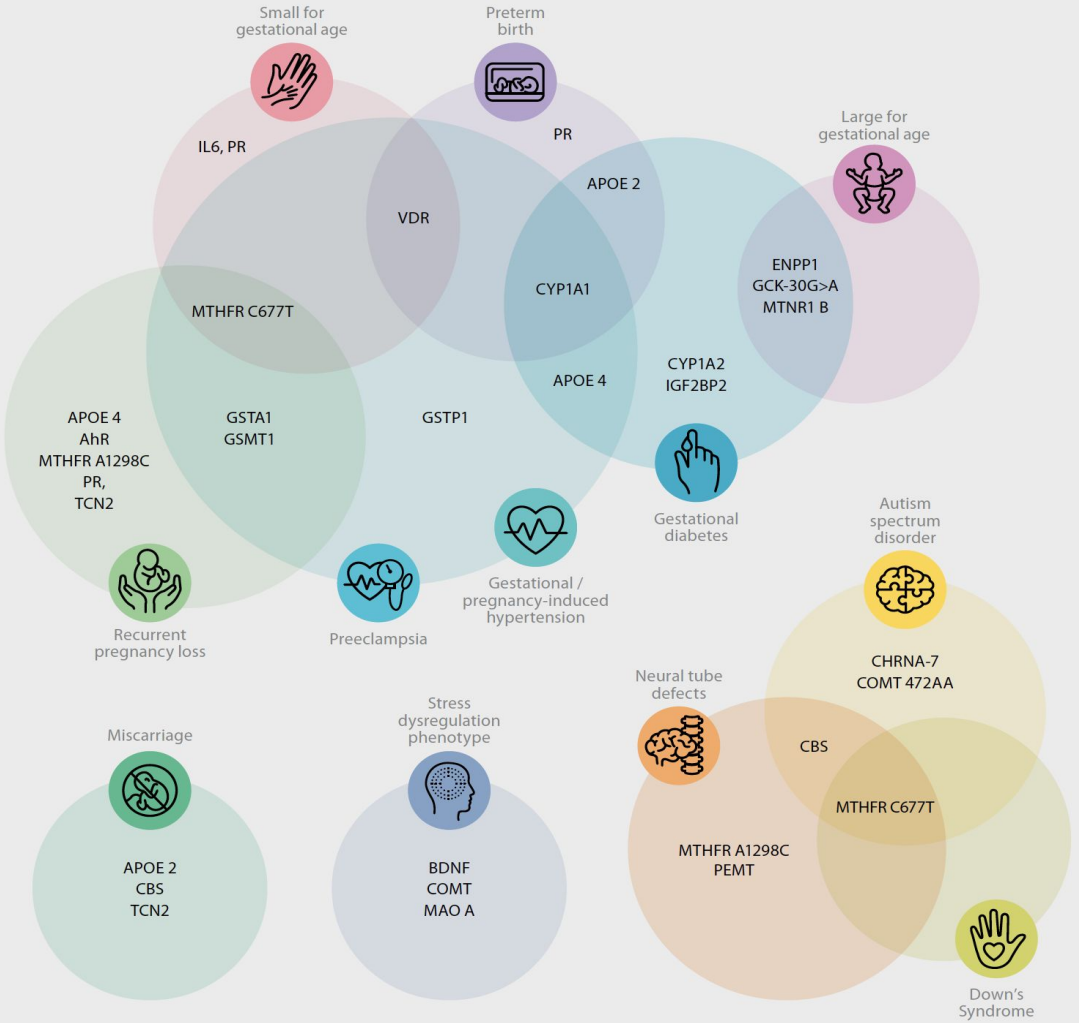
Grow Baby

Nutrigenomics:

42 SNPs in 27 genes involved in 11 key biological processes associated with maternal and fetal health outcomes.



GrowBaby®


Small for gestational age
Preterm birth
IL6, PR
PR
Large for gestational age
APOE 2
VDR
ENPP1
GCK-30G>A
MTNR1 B
CYP1A1
MTHFR C677T
CYP1A2
IGF2BP2
APOE 4
APOE 4
AhR
MTHFR A1298C
PR,
TCN2
GSTA1
GSMT1
GSTP1
Gestational diabetes
Autism spectrum disorder
Gestational / pregnancy-induced hypertension
Recurrent pregnancy loss
Preeclampsia
CHRNA-7
COMT 472AA
Neural tube defects
Stress dysregulation phenotype
Miscarriage
CBS
MTHFR C677T
MTHFR A1298C
PEMT
APOE 2
CBS
TCN2
BDNF
COMT
MAO A
Down's Syndrome

Nutrition-Integrated Care Was Associated With Substantially Lower Rates of Adverse Outcomes

Outcome	PLUS (n=402)	SOC
Preterm birth (PTB)	**1.99%**	8.7%
Hypertensive disorders of pregnancy (HDP)	**1.00%**	4.5%
Gestational diabetes mellitus (GDM)	**0.50%**	3.7%
Small for gestational age (SGA)	**1.49%**	6.1%
Large for gestational age (LGA)	**3.48%**	9.4%

The Oregon PLUS cohort (GrowBaby) vs. Oregon standard of care (statewide comparator, March of Dimes 22):1

Nutrition-Integrated Care Was Associated With Substantially Lower Rates of Adverse Outcomes

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4.37X

LESS LIKELY TO HAVE
Pre term birth

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17.48X
LESS LIKELY TO HAVE
Gestinational diabetes

4.37X
LESS LIKELY TO HAVE
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4.37X
LESS LIKELY TO HAVE
Pre term birth

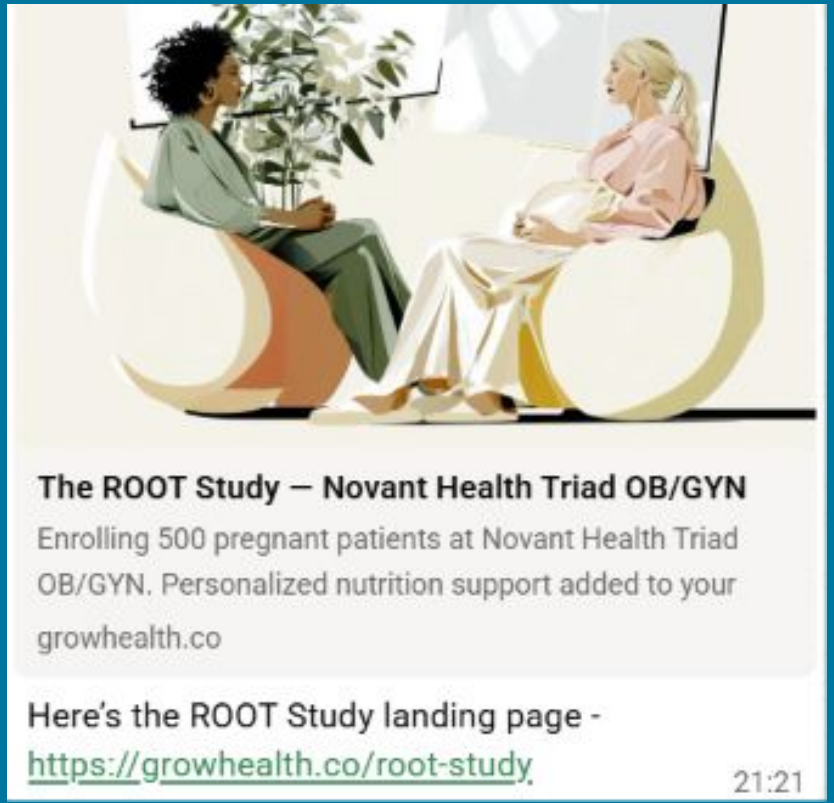
17.48X
LESS LIKELY TO HAVE
Gestinational diabetes

4.09X
LESS LIKELY TO HAVE
Babies small for gestational ageA

2.87X
LESS LIKELY TO HAVE
Babies large for gestational age

8.74X
LESS LIKELY TO HAVE
Pregnancy induce hypertension

The Oregon PLUS cohort (GrowBaby) vs. Oregon standard of care (statewide comparator, March of Dimes 22):1


The ROOT Study — Novant Health Triad OB/GYN
Enrolling 500 pregnant patients at Novant Health Triad OB/GYN. Personalized nutrition support added to your
growhealth.co
Here's the ROOT Study landing page -
https://growhealth.co/root-study
21:21

The research behind the model

The ROOT Study builds on more than fifteen years of clinical application and published research on the GrowBaby model.

2026 · Journal of Personalized Medicine

A 'Standard of Care PLUS' Model for Preterm Birth Prevention: Integrating Nutrient and Gene Variant Analysis with Targeted Interventions

Stone LP, Stone Rydbom E, Stone PM, Kim D.

2025 · Frontiers in Nutrition

Perinatal nutrition as a key regulator of genomic imprinting: A new paradigm for maternal-child health

Aronica L, Fessler SN, Stone Rydbom E, Jirtle RL.

2014 · Global Advances in Health and Medicine

Customized Nutritional Enhancement for Pregnant Women Appears to Lower Incidence of Certain Common Maternal and Neonatal Complications: An Observational Study

Stone LP, Stone PM, Rydbom EA, Stone LA, Stone TE, Wilkens LE, Reynolds K.

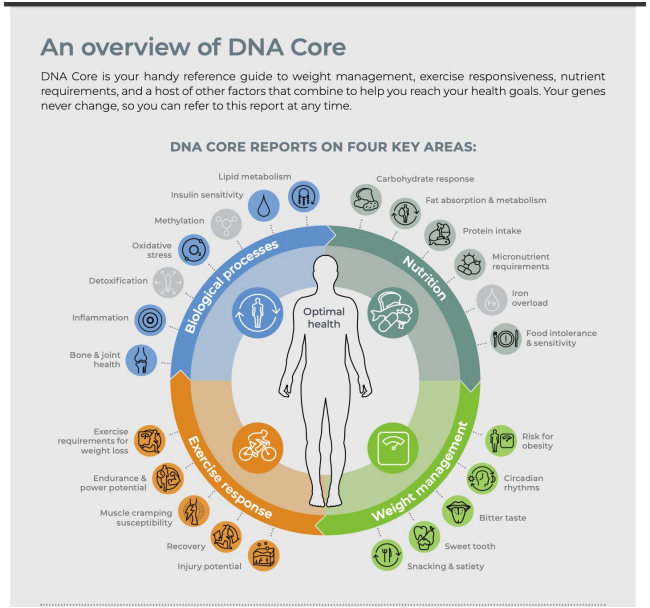
Infant Health

DNA core

Genetic profile

Personalized prevention

Targeted lifestyle advice





Genetics: Genes load the gun, the environment pulls the trigger



Foundational blueprint for understanding individual biochemical variations,

Allows clinicians to move from a one-size-fits-all approach to truly personalized interventions.

Key Information We Seek:

Pharmacogenomics: CYP450 enzyme variants affect drug metabolism

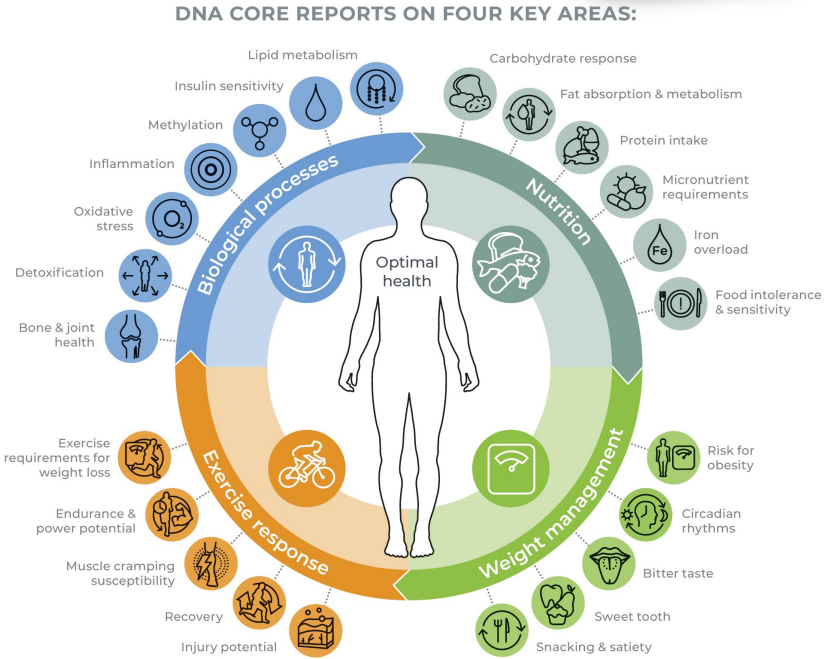
Methylation pathways: MTHFR, COMT, CBS variants affecting detoxification and neurotransmitter metabolism

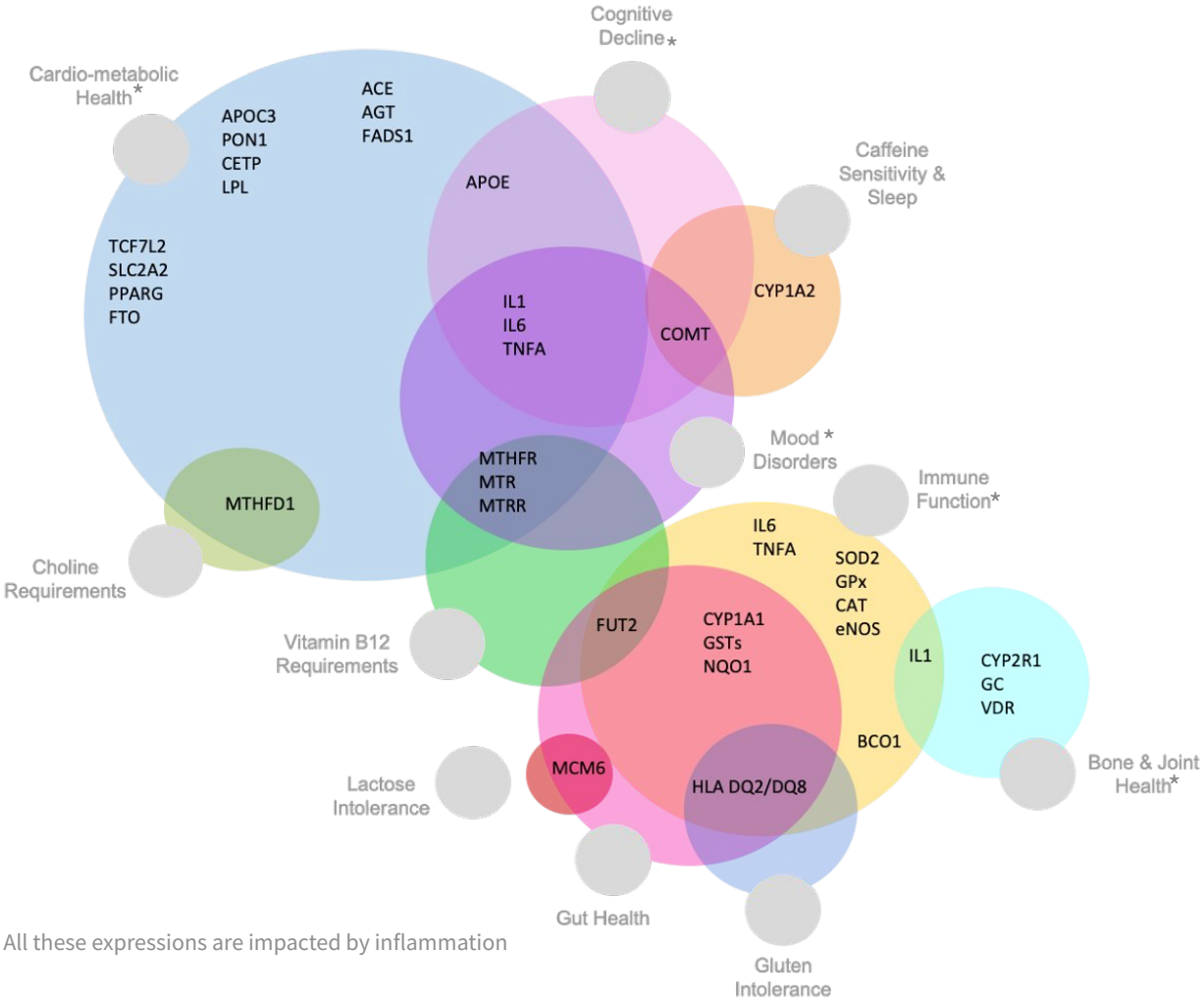
Cardiovascular risk: APOE variants, inflammatory markers like IL-6, TNF-alpha

Nutrient metabolism: Variants affecting folate, B12, vitamin D, omega-3

Detoxification capacity: Phase I & II liver detoxification enzyme variants

Inflammatory response: HLA types





* All these expressions are impacted by inflammation

dnalife®

How Genes Influence the Gut

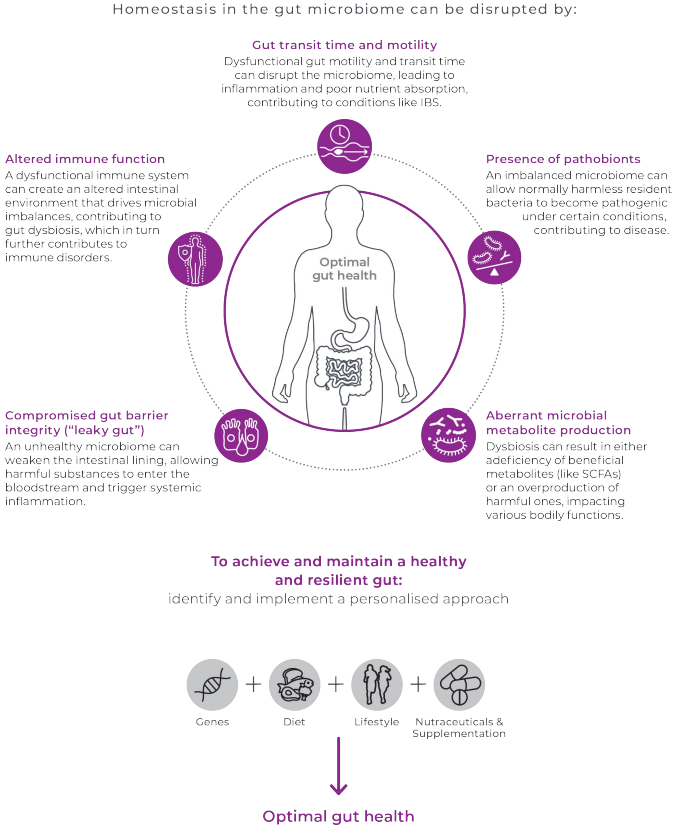
- **Host Genetics shape the "Environment":** Our genes determine the structure of our gut (e.g., gut motility, mucus production, gut barrier integrity).
- **Immune System Genes**: Genetic variations in immune response genes can influence how our body interacts with gut microbes.
- **Enzyme Production**: Genes dictate the enzymes we produce, which can affect what nutrients are available for microbes to consume.
- **HLA Genes (Human Leukocyte Antigen):** Linked to autoimmune conditions and can influence microbial recognition.
- **Gut Epithelial Barrier**: Genes play a crucial role in maintaining the integrity of the gut lining, which is vital for preventing "leaky gut."

Introduction to DNA Gut

Human/host genetic test

These alterations have negative consequences for human health and, together with genetic predisposition, increase risk for the following conditions:

- Inflammatory bowel disease (IBD), Ulcerative colitis (UC), Crohn's disease
- Irritable bowel syndrome (IBS)
- Coeliac disease or gluten sensitvity
- Lactose intolerance
- H. *pylori* infection
- Gastroenteritis
- Cognitive decline
- Depression and mood disorders



dnagut

Pediatric precision

Early Mapping

Screening for nutritional requirements and metabolic pathways from infancy to avoid developmental bottlenecks.

Nutrigenomics

Tailoring diet and supplementation to specific genetic SNPs, ensuring optimal cognitive and physical growth.

Proactive Immunity

Managing the intersection of environment and microbiome early to prevent chronic inflammatory triggers later in life.

Nordic Group

Monitoring from birth is the dream, but what is 2026 reality?

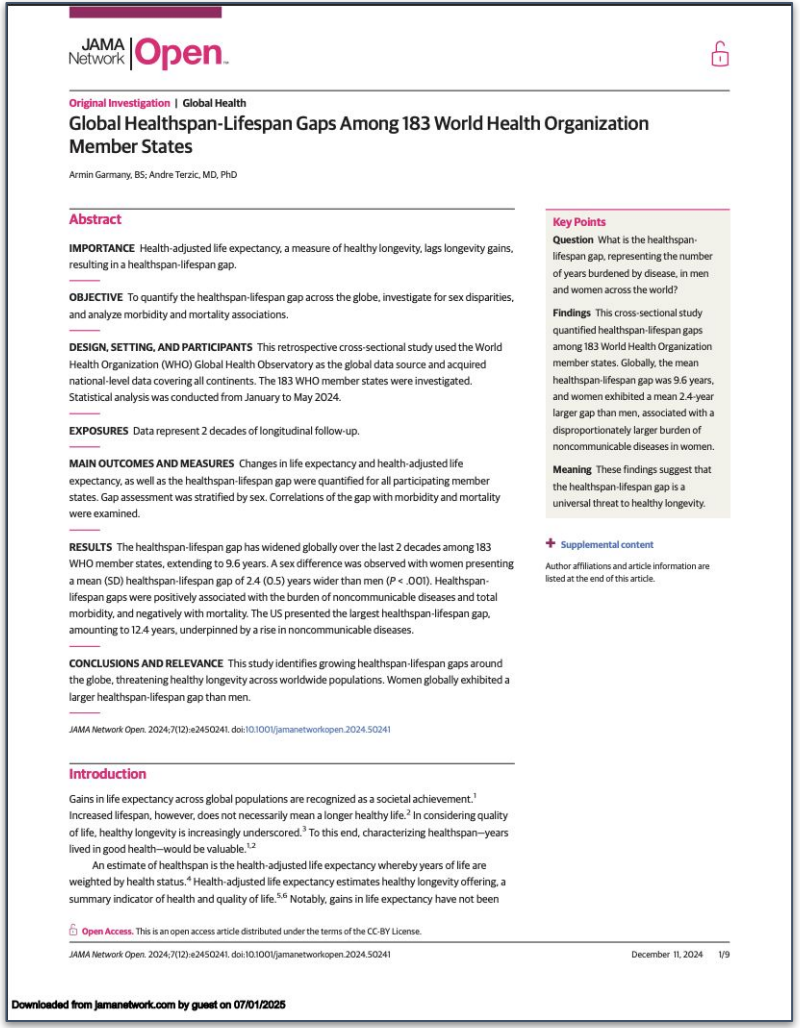
The global lifespan - healthspan gap is increasing

A cross-sectional study quantified healthspan-lifespan gaps in 183 WHO member states.

The 2024 global mean gap was 9.6 years.

The 2025 update focuses on regional disease contributions and the accelerating gap, growing at a median rate of **0.05 years** annually worldwide.

The US gap (12.4 years) is heavily driven by a disproportionate surge in non-communicable disease (NCD)



JAMA Network Open

Original Investigation | Global Health

Global Healthspan-Lifespan Gaps Among 183 World Health Organization Member States

Armin Garmany, BS; Andre Terzic, MD, PhD

Abstract

IMPORTANCE Health-adjusted life expectancy, a measure of healthy longevity, lags longevity gains, resulting in a healthspan-lifespan gap.

OBJECTIVE To quantify the healthspan-lifespan gap across the globe, investigate for sex disparities, and analyze morbidity and mortality associations.

DESIGN, SETTING, AND PARTICIPANTS This retrospective cross-sectional study used the World Health Organization (WHO) Global Health Observatory as the global data source and acquired national-level data covering all continents. The 183 WHO member states were investigated. Statistical analysis was conducted from January to May 2024.

EXPOSURES Data represent 2 decades of longitudinal follow-up.

MAIN OUTCOMES AND MEASURES Changes in life expectancy and health-adjusted life expectancy, as well as the healthspan-lifespan gap were quantified for all participating member states. Gap assessment was stratified by sex. Correlations of the gap with morbidity and mortality were examined.

RESULTS The healthspan-lifespan gap has widened globally over the last 2 decades among 183 WHO member states, extending to 9.6 years. A sex difference was observed with women presenting a mean (SD) healthspan-lifespan gap of 2.4 (0.5) years wider than men ($P < .001$). Healthspan-lifespan gaps were positively associated with the burden of noncommunicable diseases and total morbidity, and negatively with mortality. The US presented the largest healthspan-lifespan gap, amounting to 12.4 years, underpinned by a rise in noncommunicable diseases.

CONCLUSIONS AND RELEVANCE This study identifies growing healthspan-lifespan gaps around the globe, threatening healthy longevity across worldwide populations. Women globally exhibited a larger healthspan-lifespan gap than men.

JAMA Network Open. 2024;7(12):e2450241. doi:10.1001/jamanetworkopen.2024.50241

Key Points

Question What is the healthspan-lifespan gap, representing the number of years burdened by disease, in men and women across the world?

Findings This cross-sectional study quantified healthspan-lifespan gaps among 183 World Health Organization member states. Globally, the mean healthspan-lifespan gap was 9.6 years, and women exhibited a mean 2.4-year larger gap than men, associated with a disproportionately larger burden of noncommunicable diseases in women.

Meaning These findings suggest that the healthspan-lifespan gap is a universal threat to healthy longevity.

Supplemental content

Author affiliations and article information are listed at the end of this article.

Introduction

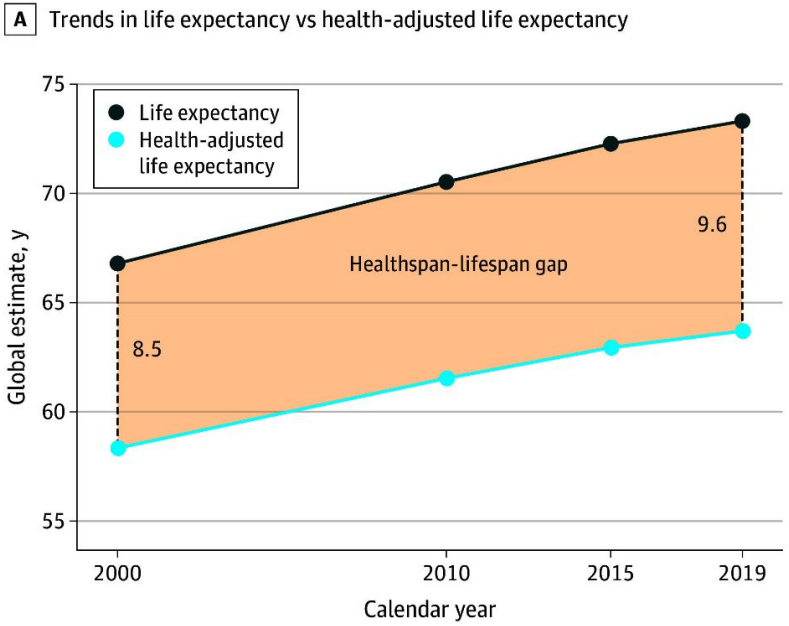
Gains in life expectancy across global populations are recognized as a societal achievement.[1] Increased lifespan, however, does not necessarily mean a longer healthy life.[2] In considering quality of life, healthy longevity is increasingly underscored.[3] To this end, characterizing healthspan—years lived in good health—would be valuable.[1,2]

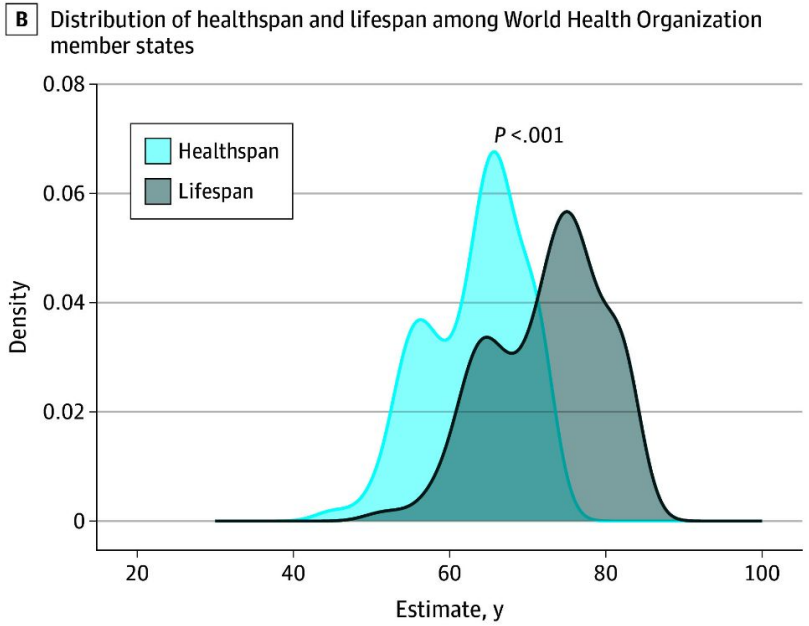
An estimate of healthspan is the health-adjusted life expectancy whereby years of life are weighted by health status.[4] Health-adjusted life expectancy estimates healthy longevity offering, a summary indicator of health and quality of life.[5,6] Notably, gains in life expectancy have not been

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The HealthSpan Challenge





Source: Garmany, A. and Terzic, A., 2024. Global healthspan-lifespan gaps among 183 World Health Organization member states. *JAMA Network Open*, 7(12).

The Longevity Dividend

Developments in life expectancy and the growing emphasis on biological and 'healthy' aging raise a number of important questions for health scientists and economists alike. Is it preferable to make lives healthier by compressing morbidity, or longer by extending life?



ANALYSIS
https://doi.org/10.1038/s43587-021-00080-0
nature aging
Check for updates

OPEN

The economic value of targeting aging

Andrew J. Scott[1]✉, Martin Ellison[2] and David A. Sinclair[3]

Developments in life expectancy and the growing emphasis on biological and 'healthy' aging raise a number of important questions for health scientists and economists alike. Is it preferable to make lives healthier by compressing morbidity, or longer by extending life? What are the gains from targeting aging itself compared to efforts to eradicate specific diseases? Here we analyze existing data to evaluate the economic value of increases in life expectancy, improvements in health and treatments that target aging. We show that a compression of morbidity that improves health is more valuable than further increases in life expectancy, and that targeting aging offers potentially larger economic gains than eradicating individual diseases. We show that a slowdown in aging that increases life expectancy by 1 year is worth US$38 trillion, and by 10 years, US$367 trillion. Ultimately, the more progress that is made in improving how we age, the greater the value of further improvements.

Life expectancy (LE) has increased dramatically over the past 150 years[1]. Analysis of the Global Burden of Disease dataset[2] suggests that the proportion of life in good health has remained broadly constant, implying increasing years in poor health. Furthermore, the disease burden is shifting towards chronic non-communicable diseases, estimated to have caused 72.3% of deaths in the United States in 2016. The result is "a substantial part of life, and certainly most deaths, now occur in a period in the lifespan when the risk for frailty and disability increases exponentially."[3] As a consequence, there is a growing emphasis on 'healthy aging' and an emerging body of research focusing on the biology of aging (see refs. [4,5]). According to another paper, "this era marks an inflection point, not only in aging research but also for all biological research that affects the human healthspan."[6]

These developments pose a number of important questions. Is it preferable to make lives healthier by compressing morbidity, or longer by extending life? What are the gains from targeting aging itself, with its potential to make lives both healthier and longer? How does the value of treating aging compare to eradicating specific diseases? How will the economic value of these gains evolve over time? To answer these questions, we take an economic rather than biological perspective. Specifically, we use the value of statistical life (VSL) methodology to place a monetary value on the gains from longer life, better health, and changes in the rate at which we age[7–9].

VSL models have two distinct advantages for our purposes. First, they are already used by government agencies to evaluate different policy measures and treatments, for example[10,11]. Second, as our model is based around optimizing economic agents, we can calculate not only the current gains from targeting aging but also how these gains will evolve in response to potential future changes in health and LE. The results reveal a distinctive feature of age-targeting treatments. Interactions between health, longevity, economic decisions and demographics create a virtuous circle, such that the more successful society is in improving how we age, the greater the economic value of further improvements.

Results

Our economic model is based on that in ref. [9] and calibrated to current US data. In the model, individuals make choices about consumption, hours worked and leisure based on wage rates, interest rates, retirement age, and knowledge of remaining LE and likely future health. Changes in health or longevity lead to changes in these economic decisions, enabling us to estimate an individual's willingness to pay (WTP) for these improvements. WTP is measured in US dollars and reflects the increase in VSL induced by improvements in health and longevity. VSL is the sum of the value of each remaining year of life, discounted to the present day and weighted by the survival rate. As the value of each year of life depends on health, consumption and leisure, the VSL incorporates both the quantity and quality of expected life remaining. Importantly, this means that the VSL is higher than an individual's lifetime income; life is valuable in its own right because individuals value time, health and leisure.

The demographic data underpinning our analysis are: (1) a survival function[12], in which mortality risk increases exponentially with age; (2) a health deficit function[12], which also increases exponentially with age; and (3) the 2017 population structure and birth rates from the US Census Bureau. In our baseline calculations, LE and healthy life expectancy (HLE) at birth are 78.9 and 68.5 years, respectively, based on current US data. Following ref. [14], we set the average VSL of an adult aged between 25 and 65 years to US$11.5 million. Although our dollar WTP values are sensitive to the precise calibration of our model, the relative importance of different treatments for aging is not.

Life extension (the Struldbrugg case). We first focus on improving LE which, with reference to *Gulliver's Travels*[15], we refer to as the Struldbrugg case. Struldbruggs, both male and female, are born immortal but age normally, so live in continuously worsening health. In our simulations, we achieve this by reducing the rate at which mortality increases with age while holding unchanged the rate at which health declines. The result is an expansion of morbidity such that the ratio of HLE to LE deteriorates.

The WTP for LE increases depends on which years benefit from lower mortality. To provide consistency across simulations, we assume mortality is subject to a compensating effect[16] whereby it reaches a rate M at age T. Under this specification, there are two ways to extend LE. The first is to rectangularize the survival function such that M and T are kept constant but mortality falls at all ages less than T while rising more rapidly at T (Fig. 1, blue survival function). The second involves increasing lifespan, such that mortality reaches M at higher values of T. In this case, survival rates

[1]Department of Economics, London Business School, London, UK. [2]Department of Economics, University of Oxford, Oxford, UK. [3]Harvard Medical School, Boston, MA, USA. ✉e-mail: ascott@london.edu

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Now let’s imagine we’re a 120 year old in 2126. What would we say to our 20 year old self?

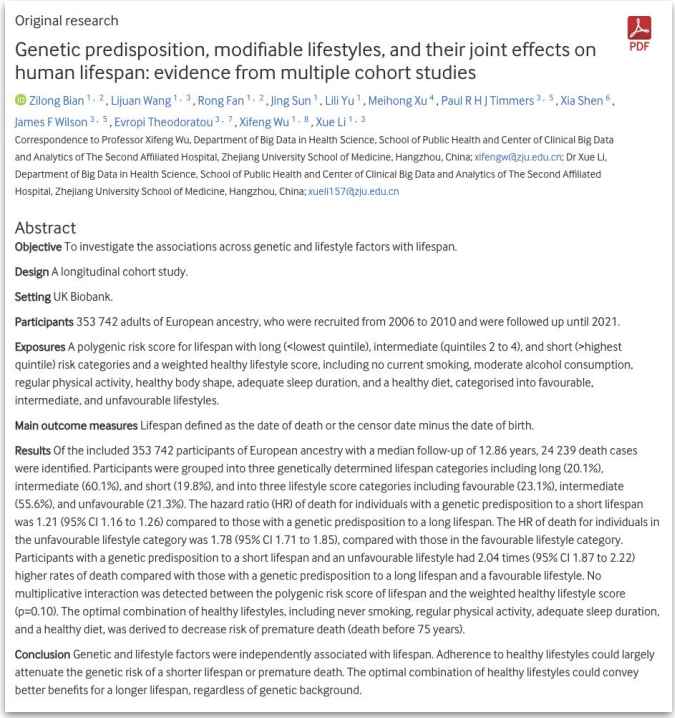
Bill Clinton, 26 June 2000

“We are here to celebrate the completion of the first survey of the entire human genome. Without a doubt, this is the most important, most wondrous map ever produced by human kind.”



To demonstrate the importance of understanding our genetic makeup

Genetic predisposition, **modifiable lifestyles**, and their **joint** effects on human **lifespan**: evidence from **multiple** cohort studies



Original research

PDF

Genetic predisposition, modifiable lifestyles, and their joint effects on human lifespan: evidence from multiple cohort studies

Zilong Bian [1, 2], Lijuan Wang [1, 3], Rong Fan [1, 2], Jing Sun [1], Lili Yu [1], Meihong Xu [4], Paul R H J Timmers [3, 5], Xia Shen [6], James F Wilson [3, 5], Evropi Theodoratou [3, 7], Xifeng Wu [1, 8], Xue Li [1, 3]

Correspondence to Professor Xifeng Wu, Department of Big Data in Health Science, School of Public Health and Center of Clinical Big Data and Analytics of The Second Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China; xifengwu@zju.edu.cn; Dr Xue Li, Department of Big Data in Health Science, School of Public Health and Center of Clinical Big Data and Analytics of The Second Affiliated Hospital, Zhejiang University School of Medicine, Hangzhou, China; xueli157@zju.edu.cn

Abstract

Objective To investigate the associations across genetic and lifestyle factors with lifespan.

Design A longitudinal cohort study.

Setting UK Biobank.

Participants 353 742 adults of European ancestry, who were recruited from 2006 to 2010 and were followed up until 2021.

Exposures A polygenic risk score for lifespan with long (<lowest quintile), intermediate (quintiles 2 to 4), and short (>highest quintile) risk categories and a weighted healthy lifestyle score, including no current smoking, moderate alcohol consumption, regular physical activity, healthy body shape, adequate sleep duration, and a healthy diet, categorised into favourable, intermediate, and unfavourable lifestyles.

Main outcome measures Lifespan defined as the date of death or the censor date minus the date of birth.

Results Of the included 353 742 participants of European ancestry with a median follow-up of 12.86 years, 24 239 death cases were identified. Participants were grouped into three genetically determined lifespan categories including long (20.1%), intermediate (60.1%), and short (19.8%), and into three lifestyle score categories including favourable (23.1%), intermediate (55.6%), and unfavourable (21.3%). The hazard ratio (HR) of death for individuals with a genetic predisposition to a short lifespan was 1.21 (95% CI 1.16 to 1.26) compared to those with a genetic predisposition to a long lifespan. The HR of death for individuals in the unfavourable lifestyle category was 1.78 (95% CI 1.71 to 1.85), compared with those in the favourable lifestyle category. Participants with a genetic predisposition to a short lifespan and an unfavourable lifestyle had 2.04 times (95% CI 1.87 to 2.22) higher rates of death compared with those with a genetic predisposition to a long lifespan and a favourable lifestyle. No multiplicative interaction was detected between the polygenic risk score of lifespan and the weighted healthy lifestyle score (p=0.10). The optimal combination of healthy lifestyles, including never smoking, regular physical activity, adequate sleep duration, and a healthy diet, was derived to decrease risk of premature death (death before 75 years).

Conclusion Genetic and lifestyle factors were independently associated with lifespan. Adherence to healthy lifestyles could largely attenuate the genetic risk of a shorter lifespan or premature death. The optimal combination of healthy lifestyles could convey better benefits for a longer lifespan, regardless of genetic background.

Source: Bian et al. BMJ Evidence-Based Medicine Month 2024

Genetic predisposition, modifiable lifestyles, and their joint effects on human lifespan: evidence from multiple cohort studies

To investigate the associations across genetic and lifestyle factors with lifespan using a longitudinal cohort study.

Exposure:

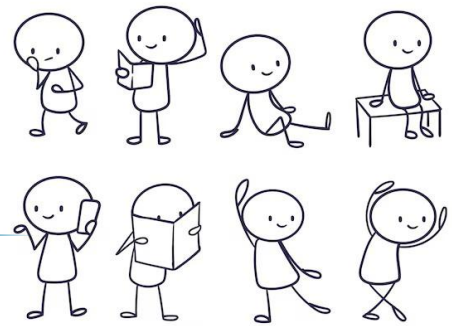
Lifespan polygenic risk score

- ☐ Short
- ☐ Intermediate
- ☐ Long

+

Healthy lifestyle score: smoking, alcohol, physical activity, body shape, healthy diet

- ✔ Favourable
- ✔ Intermediate
- ✔ Unfavourable



UK Biobank sample: 353,742 participants recruited from 2006 to 2010 & followed up (F/U) to 2021. Median F/U: 12.86 years, 24 239 death cases were identified

Outcome:
Lifespan defined as date of death.

Source: Bian et al. BMJ Evidence-Based Medicine Month 2024

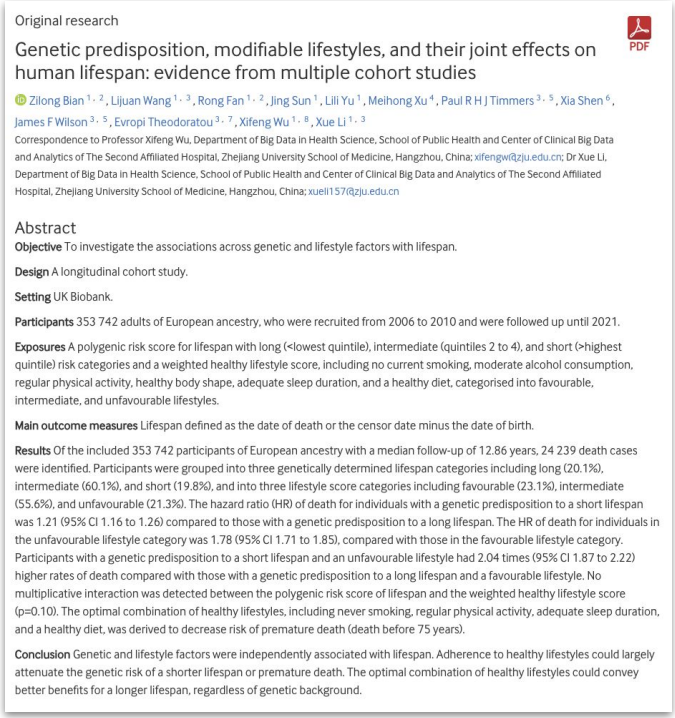
Genetic predisposition, modifiable lifestyles, and their joint effects on human lifespan: evidence from multiple cohort studies

"The subgroup of long lifespan (genes) and unfavourable lifestyle has the most years of life lost

The genetic predisposition to a shorter lifespan can be substantially compensated by having a healthy lifestyle.

Participants with high genetic risk could prolong life by approx 5 years (if they change) at age 40 to a favourable lifestyle."

It's not so much our genes that make the difference. It's knowing we have those genes and changing our behaviour around them.



Original research

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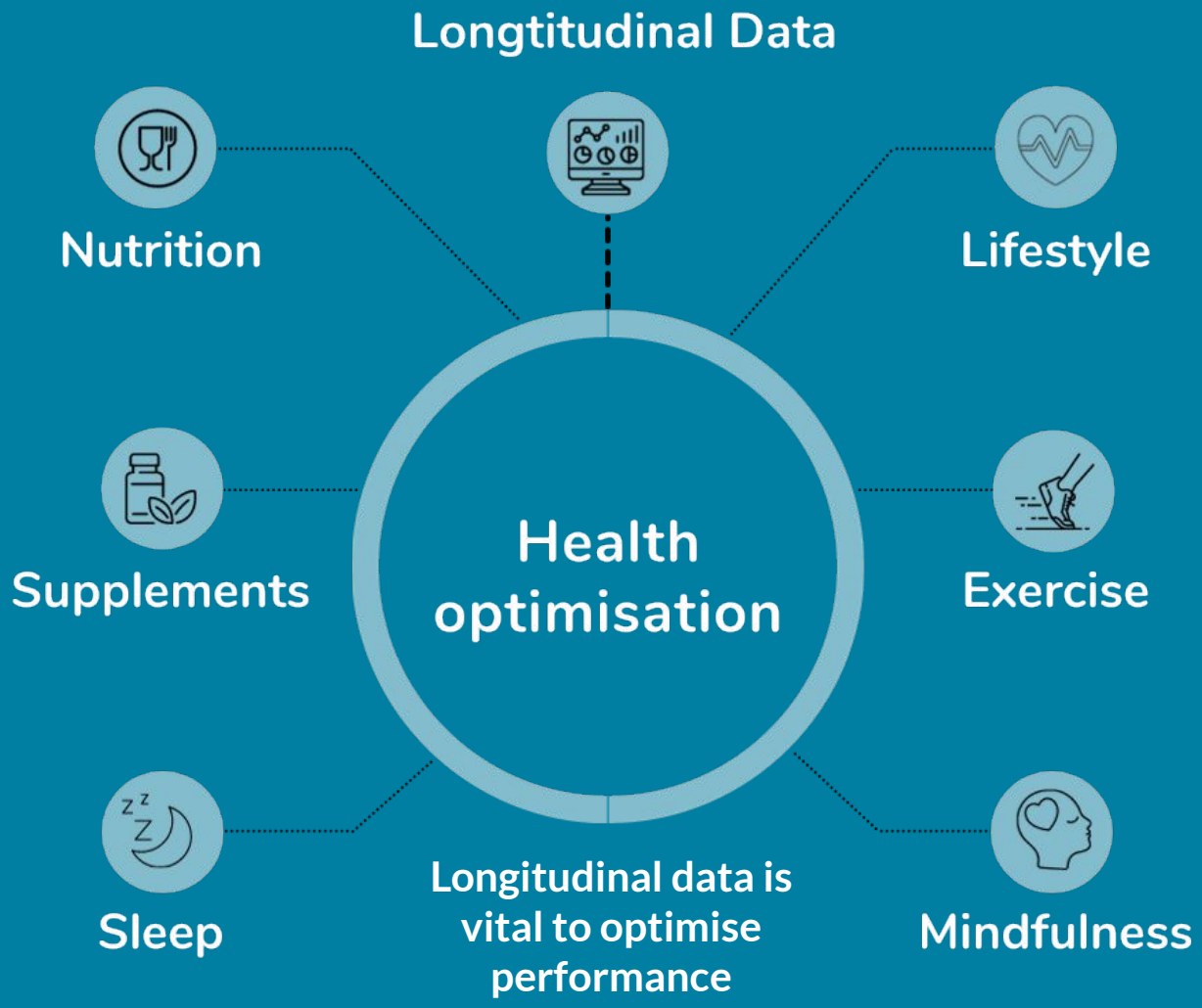
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Source: Bian et al. BMJ Evidence-Based Medicine Month 2024


Longtitudinal Data
Nutrition
Lifestyle
Supplements
Health optimisation
Exercise
Sleep
Longitudinal data is vital to optimise performance
Mindfulness

The biohacker toolkit

30

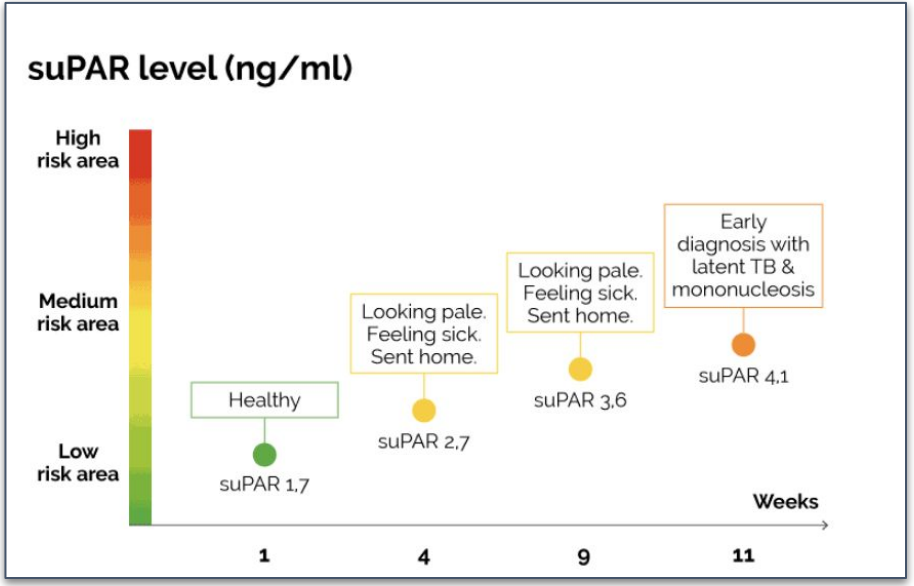
Years of personal data

My Hacks

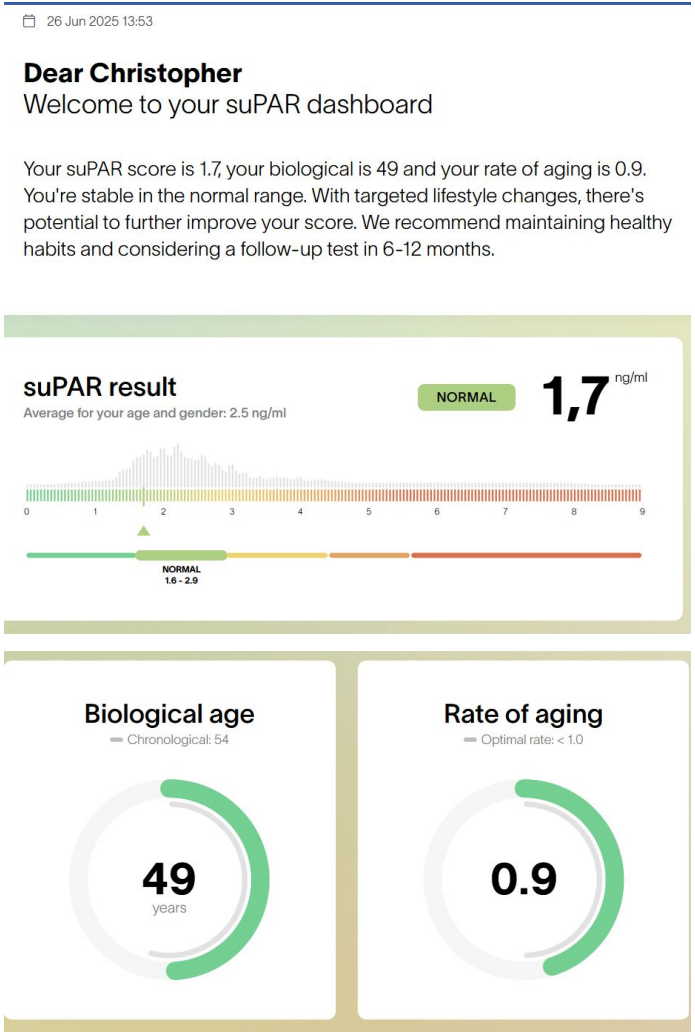
SuPar: The Inflammatory Thermometer

We can measure underlying inflammatory state.

The suPAR test, provided by ViroGates, is an advanced biomarker that provides a stable, long-term measure of systemic inflammation.



suPAR is a test that measures underlying inflammation, developed by researchers in Denmark.



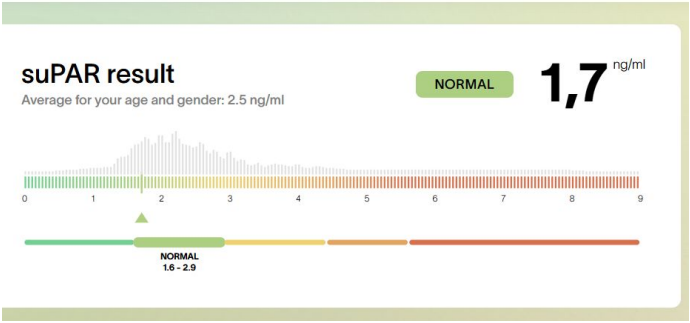
“Look at your biometrics!
You’re so lucky”

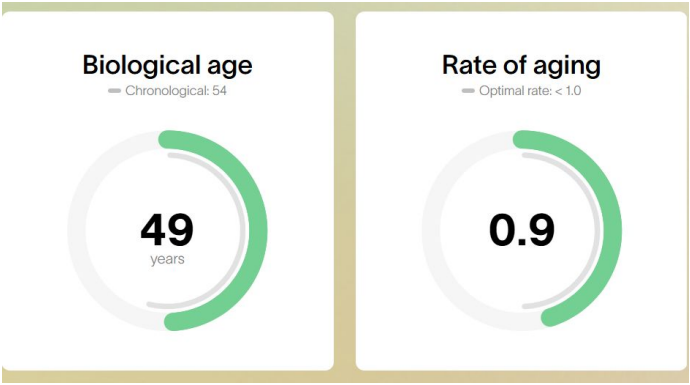
My score did not happen by luck! I have been optimising my health since 1994.

26 Jun 2025 13:53

Dear Christopher
Welcome to your suPAR dashboard

Your suPAR score is 1.7, your biological is 49 and your rate of aging is 0.9. You're stable in the normal range. With targeted lifestyle changes, there's potential to further improve your score. We recommend maintaining healthy habits and considering a follow-up test in 6-12 months.





Enhancing healthspan (& longevity) **isn’t luck…**

…It’s preparation

I'm 55 years old
As CEO of Nordic Group, a global company. Where I travel 200+ days a year

I have to be fit and healthy to run the business

I have been optimising my health since 1994.










BPC 157
with Arginate Salt
Dietary Supplement







Latest hacks and interventions

Hyperbaric chamber for improved oxygen delivery

Cold immersion for metabolic and immune benefits

Ozone therapy for enhanced cellular function

Weber intravenous laser for targeted cellular repair

Hypoxia training for cardiorespiratory adaptation

Peptides and IV vitamins for vital nutrients

A-pheresis

Data = power

Singular data on its own
is not enough.

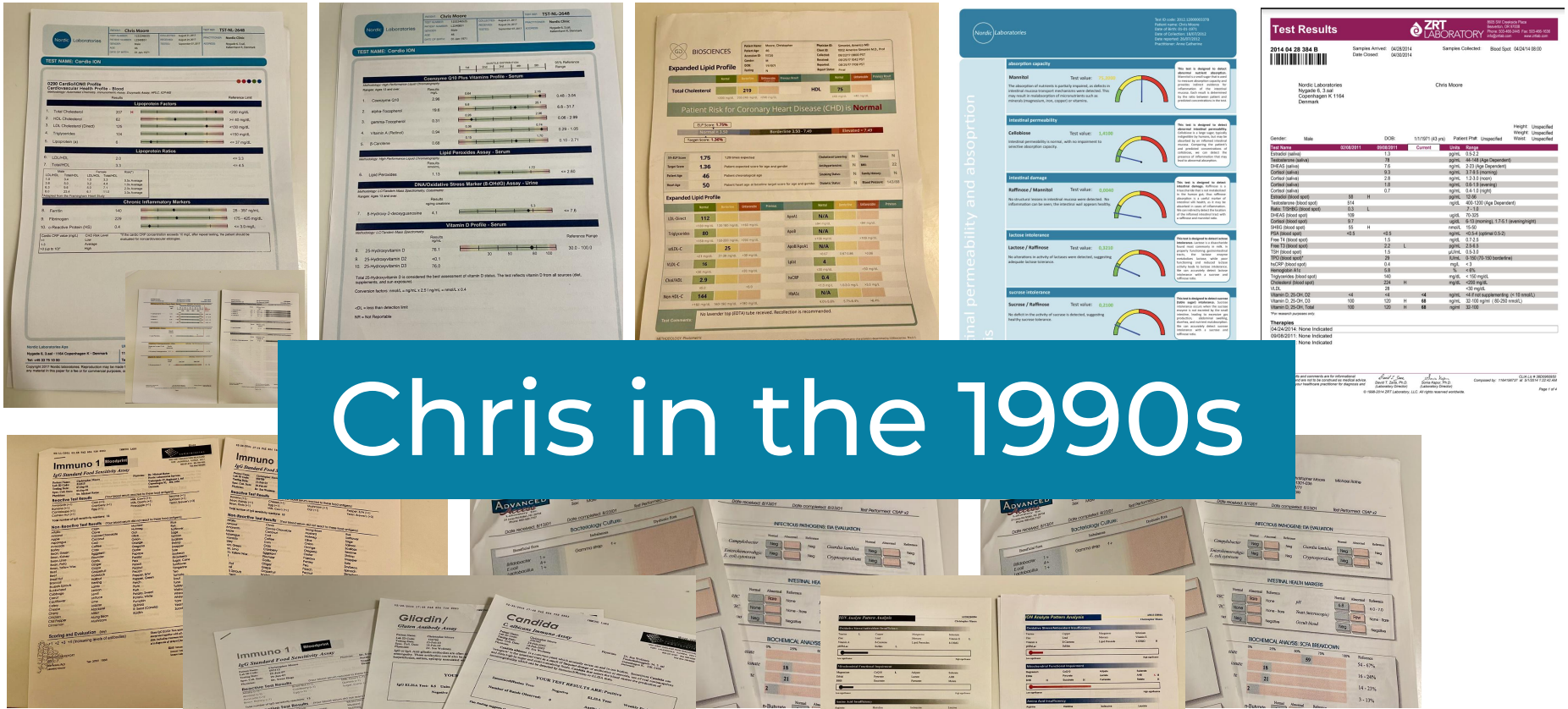
We need multiple data points to get a better understanding of how our body works.

105



This is one of the first pieces of data I found out about myself.

1995, singular data.
Starting point.



Chris in the 1990s

Gluten Intolerance

Particular nutrients and certain food components in different foodstuffs can affect in ways. With new research coming to light in this area, specific genes can be tested to how an individual might respond to a particular food component.

	Gene Name	Genetic Variation	Your Result
Gluten intolerance	HLA	DQ2/DQ8	DQ8

Nordic Laboratories

PATIENT NUMBER: N/A
GENDER: Male
AGE: 46
DATE OF BIRTH: 01-Jan-1971

PRACTITIONER: Nordic Clinic
ADDRESS: Nygade 6, 3.sal, København K, Denmark

TEST NAME: Oxidised LDL & Liposcan Panel

Test	Result		Unit	Reference Value
CLINICAL / Anamnestic data				
Anamnesis	none			
Fat Metabolism (SI units)				
Cholesterol	5.64	+	mmol/l	< 5.17
Triglycerides	1.19		mmol/l	< 1.7
LIPOSCAN (SI units)				
HDL	1.34		mmol/l	> 1.03
LDL	3.34		mmol/l	< 3.36
LDL/HDL	2.5			< 3
VLDL	0.98	+	mmol/l	< 0.59
	0.83		mmol/l	< 1.66
Lipoprotein fractions (SI units)				

TEST NAME: Complete Thy

Test Name	02/08/2
Estradiol (Saliva)	
Testosterone (Saliva)	
DHEAS (Saliva)	
Cortisol (Saliva)	
Cortisol (Saliva)	
Cortisol (Saliva)	
Cortisol (Saliva)	
Estradiol (Blood Spot)	58
Testosterone (Blood Spot)	514
Ratio: T/SHBG (Blood Spot)	0.3
DHEAS (Blood Spot)	109
SHBG (Blood Spot)	55

DATE OF BIRTH: 01-Jan-1971

TEST NAME: Celiac & Gluten Sensitivity (Serum)

Celiac & Gluten Sensitivi

Tissue Transglutaminase (tTG) IgA			
Tissue Transglutaminase (tTG) IgG			
Deamidated Gliadin Peptide (DGP) IgA	< 5.2	U	< 20.0
Deamidated Gliadin Peptide (DGP) IgG	< 2.8	U	< 20.0
Gliadin IgA	8.0	U	< 20.0
Gliadin IgG	4.0	U	< 20.0

Chris in the 2020s

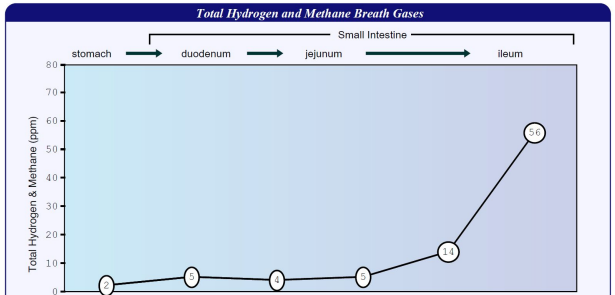
Additional Dysbiotic/Overgrowth Bacteria

	Result	
Bacillus spp.	6.26e5	High
Enterococcus faecalis	1.79e3	
ococcus faecium	6.17e2	
anella spp.	<dl	
domonas spp.	<dl	
domonas aeruginosa	<dl	
ylococcus spp.	<dl	
ylococcus aureus	<dl	
tococcus spp.	1.64e4	High
anobacteriaceae (family)	1.65e8	

ential Autoimmune Triggers

	Result
bacter spp.	<dl
bacter freundii	<dl
iella spp.	<dl
iella pneumoniae	<dl
ium subsp. paratuberculosis	<dl

Bacterial Overgrowth of the Small Intestine
Breath Test



gG - BLOODSPOT - ENGLISH

TEST	SCORE	CLASS
MEAT & POULTRY		
Beef	0.147	0
Buffalo	0.168	0
Chicken	0.159	0
Duck	0.172	0
Lamb	0.158	0
Pork	0.154	0
Turkey	0.166	0
Venison	0.177	0
FISH & SHELLFISH		
Anchovy	0.175	0
Bass	0.176	0
Clam	0.162	0
Codfish	0.161	0
Crab	0.165	0
Flounder	0.188	0
Haddock	**0.242**	**1**
Halibut	0.194	0
Herring	0.175	0
Lobster	**0.215**	**1**
Mackerel	0.155	0
Mussel	0.195	0
Oyster	0.184	0
Perch	0.183	0
Red Snapper	**0.215**	**1**
Salmon	0.180	0
Scallop	0.160	0
Shrimp	0.173	0

TEST	SCORE	CLASS
VEGETABLES		
Artichoke	0.168	0
Asparagus	0.163	0
Beets	0.183	0
Bell Pepper	0.162	0
Broccoli	0.149	0
Brussel Sprouts	**0.219**	**1**
Cabbage	0.154	0
Carrot	**0.214**	**1**
Cauliflower	0.158	0
Celery	0.169	0
Cucumber	0.160	0
Eggplant	0.158	0
Garlic	0.166	0
Green Bean	0.166	0
Kale	0.162	0
Kelp	0.193	0
Lettuce	0.161	0
Mushroom	**0.230**	**1**
Okra	**0.210**	**1**
Olive, Green	0.148	0
Onion	0.160	0
Parsnip	0.169	0
Potato	0.193	0
Potato, Sweet	0.184	0
Pumpkin	0.171	0
Radish	0.183	0
Spinach	0.157	0
Squash	0.152	0

TEST	SCORE	CLASS
DAIRY & EGG		
Blue Cheese	**0.233**	**1**
Casein	**0.308**	**2**
Cheddar Cheese	**0.232**	**1**
Egg, White	0.182	0
Egg, Yolk	0.187	0
Milk, Cow's	**0.304**	**2**
Milk, Goat's	**0.232**	**1**
Milk, Sheep's	**0.214**	**1**
Mozzarella Cheese	0.190	0
Swiss Cheese	0.193	0
Whey	**0.428**	**3**
Yogurt	**0.257**	**1**
FRUITS		
Apple	0.168	0
Apricot	0.164	0
Avocado	0.154	0
Banana	0.172	0
Blackberry	0.185	0
Blueberry	0.148	0
Cantaloupe	0.157	0
Cherry	0.174	0
Coconut	0.161	0
Cranberry	0.169	0
Date	0.172	0
Fig	0.166	0
Grape	0.162	0
Grapefruit	0.155	0

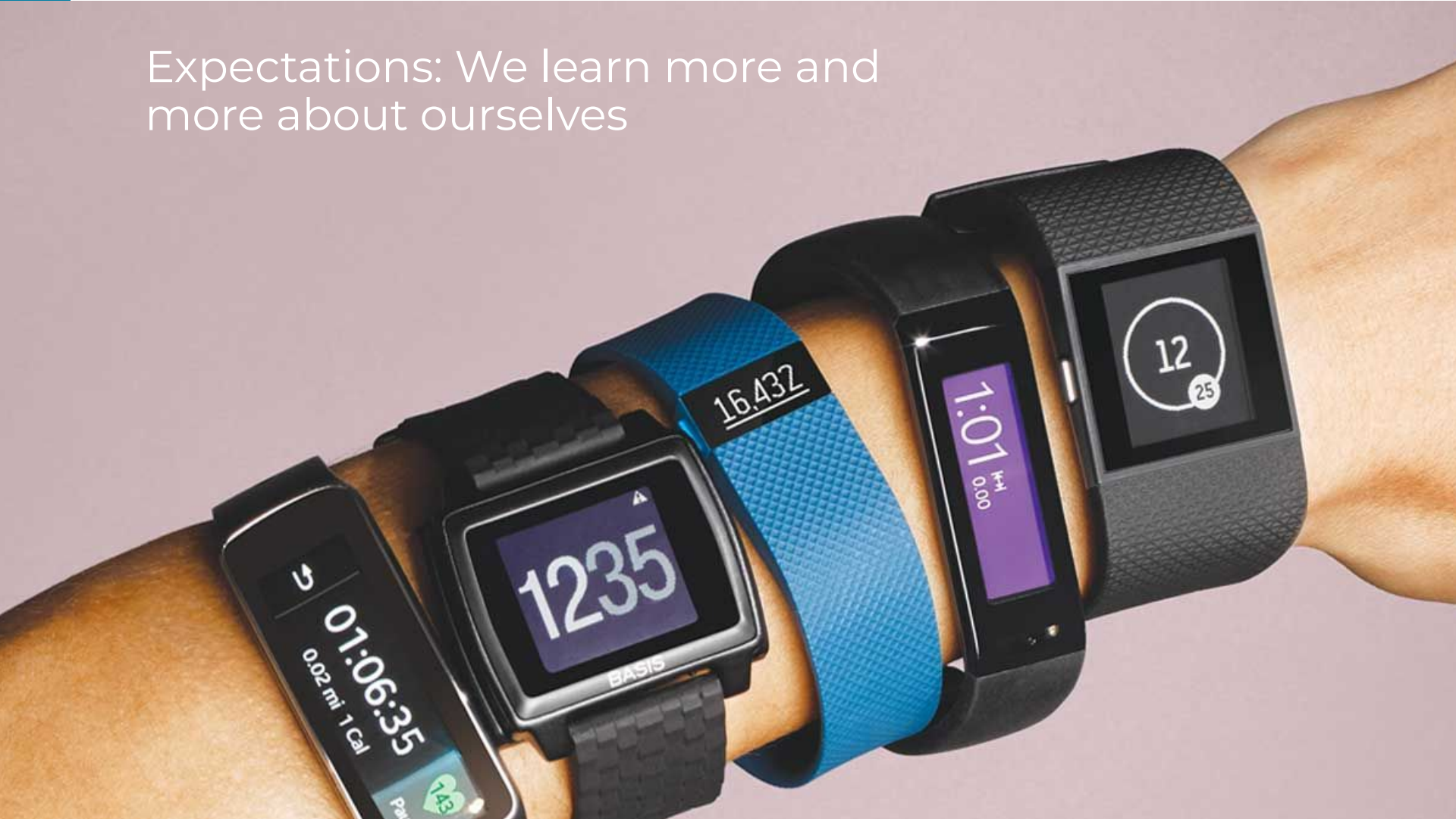
TEST
NUTS, SEE
Almond
Brazil Nut
Canola
Cashew
Chestnut
Chia Seed
Cola
Flaxseed
Hazelnut
Hemp
Macadamia
Pecan
Pine Nut
Pistachio
Poppy See
Safflower
Sesame
Sunflower S
Walnut
HERBS, SP
Basil
Black Peppe
Cilantro
Cinnamon
Cloves
Dill
Fennel See

<5.00e4
<5.00e3

It’s **NOT** just about raw data…

…it’s about
knowledge,
utilization and
implementation

Expectations: We learn more and more about ourselves

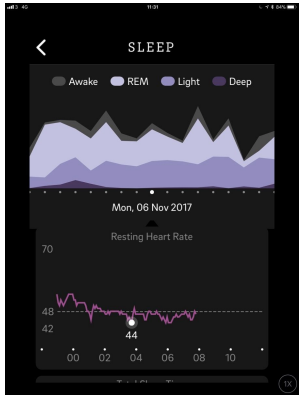


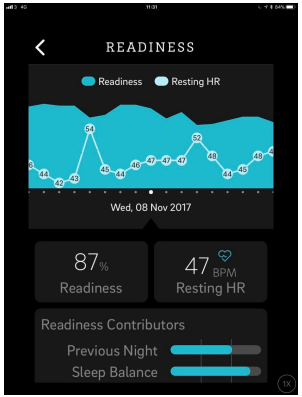


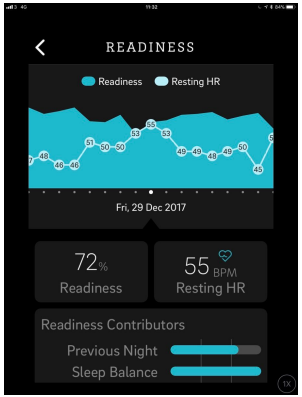
Expectations:
We learn more and more about ourselves

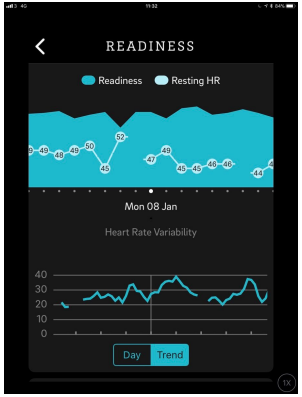
Nordic Group

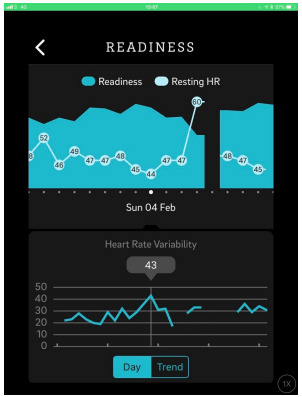
Expectations: We learn more and more about ourselves



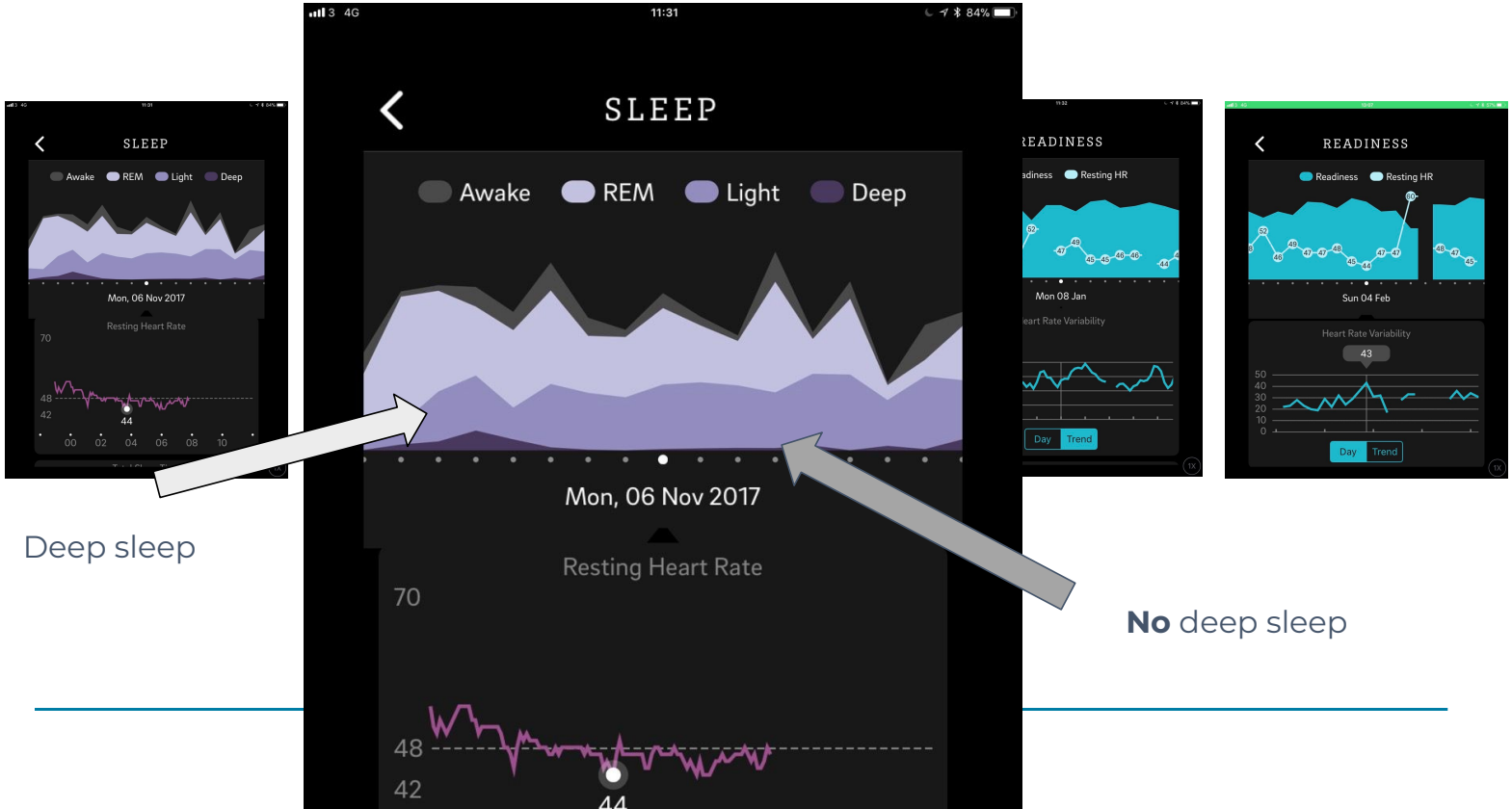








Expectations: We learn more and more about ourselves

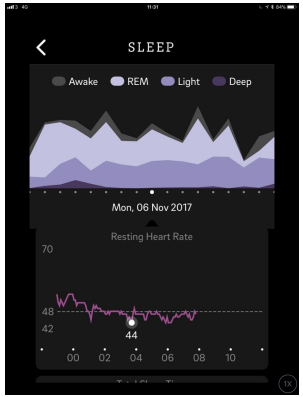


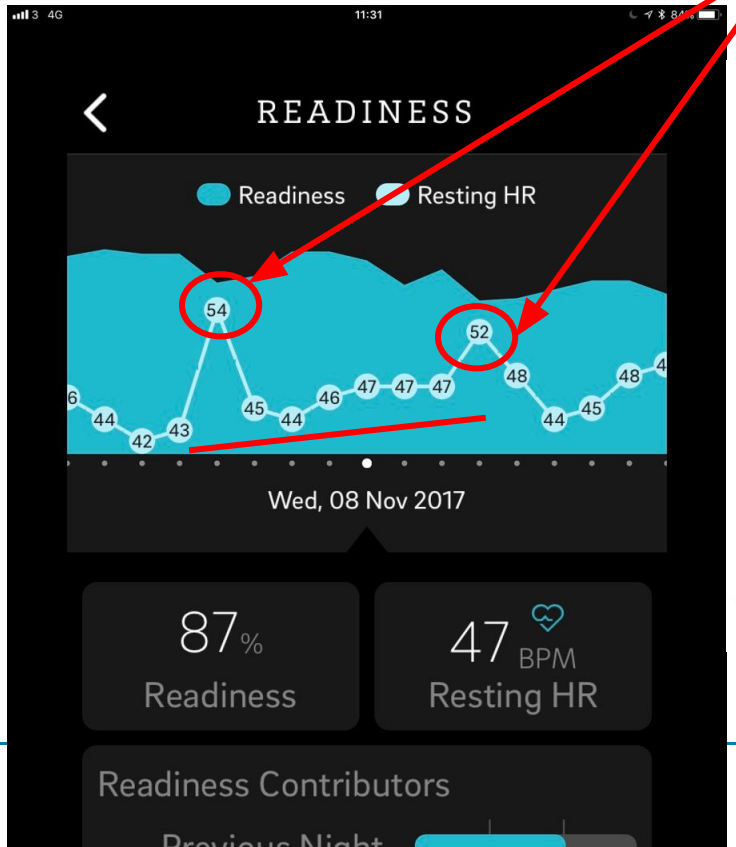
Deep sleep

No deep sleep

Resting heart rate:

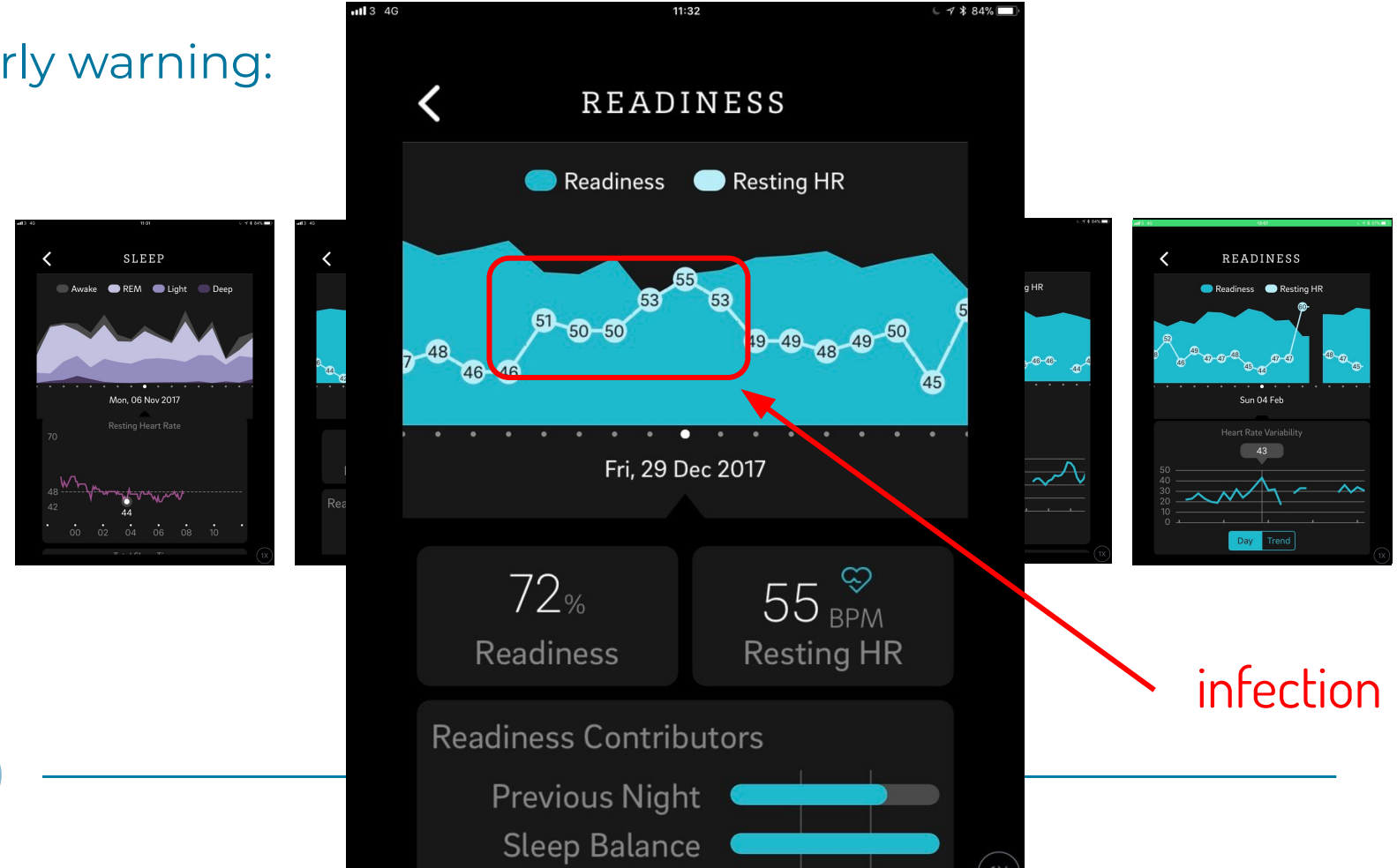
What is this due to?



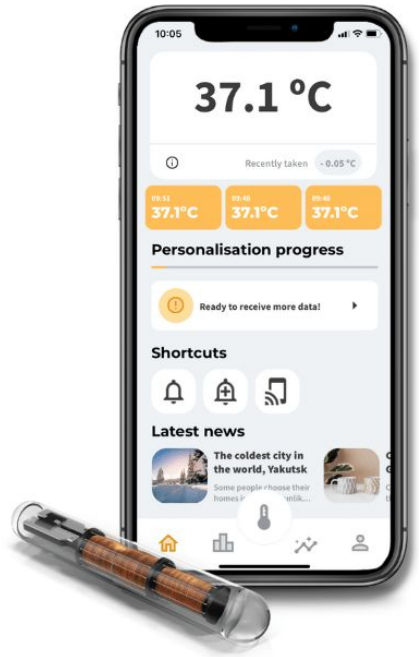




Early warning:



infection



Technology: Longitudinal data

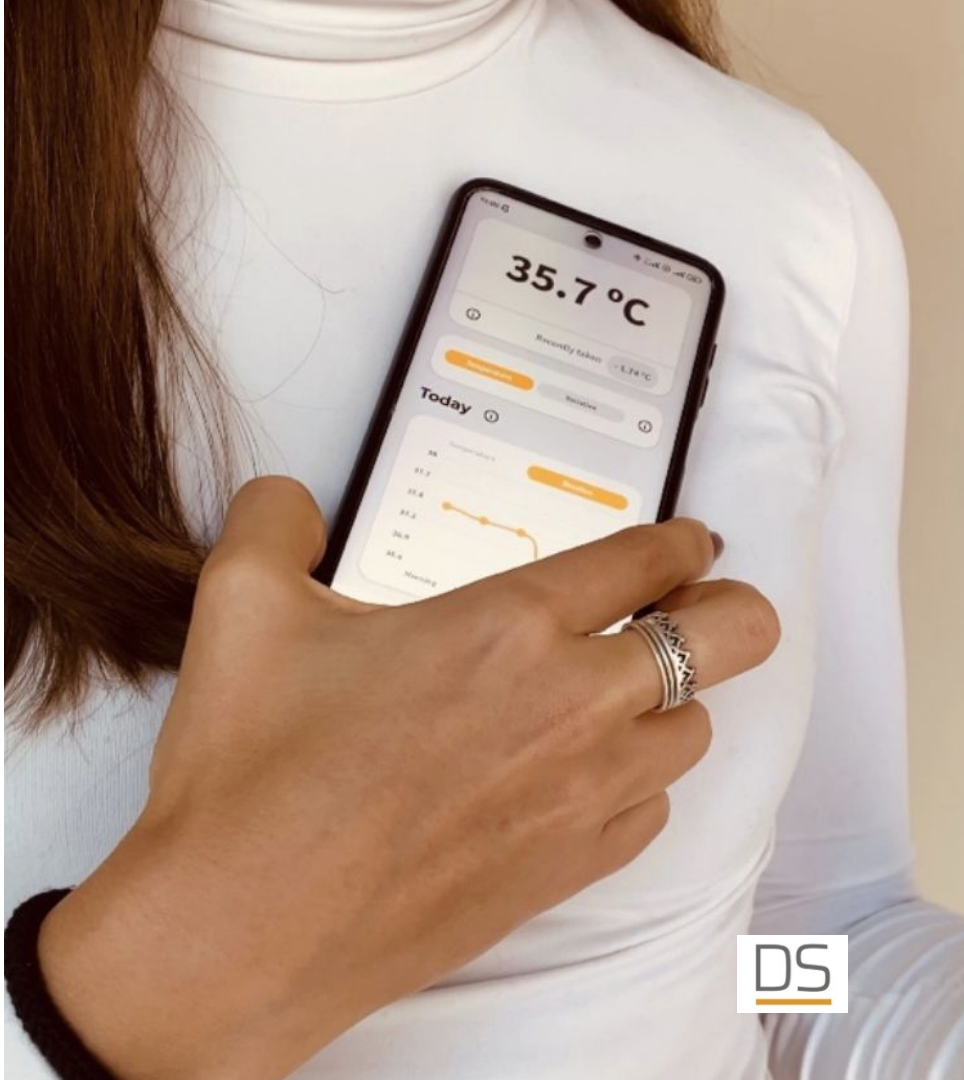
Collecting and monitoring performance data

Dsruptive DS

BeUno beta test program

The worlds' first NFC-powered temperature sensor implant.

- Sits under your skin.
- Reads to your phone
- Lasts 30 years
- Helps women with cycle management
- Helps identify infections in the body
- Allows to collect longitudinal data





Eat Organic

- Organic produce typically contains higher levels of beneficial antioxidants and polyphenols = reduced oxidative stress and inflammation.
- Organic farming practices = lower pesticide residues compared to conventional agriculture, potentially reducing long-term exposure to chemicals that may negatively impact cellular health, hormone function, and the gut microbiome.
- Soil management practices used in organic farming can lead to higher mineral content in organic foods, particularly trace minerals like zinc, iron, and magnesium, which are essential for maintaining robust immune function and cellular repair mechanisms as we age.



The Nordic Smoothie

Good nutrition every single day

Nutrition from food + supplements

Designed to provide vitamins, minerals, and micronutrients for all our body systems

It began as everyday nutrition for our family



Inuspheresis

Nordic Laboratories

PATIENT: **Chris Moore**
TEST NUMBER: T-NL-1583525 (1467786)
GENDER: Male
AGE: 54
COLLECTED: 04-Aug-2025
TESTED: 28-Aug-2025
TEST REF: **TST-NL-186373**
PRACTITIONER:
Chris Moore
test, København K, Denmark

TEST NAME: Glyphosate

Glyphosate Profile - Urine

Creatinine Value*: 188.07 mg/dl

Metabolite	Result ug/g creatinine	Reference Range (LLOQ 0.38 / 75th 1.8 / 95th 2.5)
Glyphosate	1.44	

Glyphosate is the world's most widely produced herbicide. It is a broad-spectrum herbicide that is used in more than 700 different

PATIENT: **Chris Moore**
TEST NUMBER: T-NL-1589532 (1473553)
GENDER: Male
AGE: 54
COLLECTED: 22-Aug-2025
TESTED: 08-Sep-2025
TEST REF: **TST-NL-187530**
PRACTITIONER:
Chris Moore
test, København K, Denmark

TEST NAME: Glyphosate

Glyphosate Profile - Urine

Creatinine Value*: 180.75 mg/dl

Metabolite	Result ug/g creatinine	Reference Range (LLOQ 0.38 / 75th 1.8 / 95th 2.5)
Glyphosate	<0.37	

Glyphosate is the world's most widely produced herbicide. It is a broad-spectrum herbicide that is used in more than 700 different



Inuspheresis

Effect of Inuspheresis:

Reduced Toxic metals

Reduced Glyphosate

Reduced Chemicals

Die Nordic konferenz Funktionelle Diagnostik von Kinderwunsch bis Longevity

Lernen Sie unseren Referenten kennen



Chris Moore



Corinna van der Eerden



Daniel Auer



Dr. Astrid Boeckelmann



Dr. Carola Kissel



Patrick Engels



Andschana Frobel



Dr. Carl-Peter Meschede

7. – 8. NOVEMBER 2026

DIE NORDIC KONFERENZ FRANKFURT 2026

MELDEN SIE SICH NOCH HEUTE AN!



Chris Moore,
CEO Nordic Group,
Denmark

End comment:

Integrate genetic and phenotype data. Add good nutrition. Review data sets. Re test continually. Will help people not just live longer, but live better.
#healthcareforthefuture

cm@nordicgroup.eu

YouTube Nordic Laboratories & DNAlife

LinkedIn www.linkedin.com/in/chris-moore-a112b91